



Designing External Pacing Pads to Minimize Patient Pain and Discomfort

Group 4967

Department of Bioengineering

Imperial College London

Supervisor:
Dr Nick Linton

April, 2026

Designing External Pacing Pads to Minimize Patient Pain and Discomfort

Vinayak Gupta, Michelle Liao, Sean Obi, Vivian Sullyunna,
Rohan Turner, Jaehoon Yi, Richard Zhao

April 15, 2026

Abstract

External pacing is a life-saving intervention for patients with cardiac arrhythmias, particularly bradycardia and heart block. It involves delivering electrical current across the chest wall via two surface electrodes to stimulate cardiac activity, establishing a normal rhythm. However, as electrical stimulation crosses the chest wall, it can activate skeletal muscle, causing painful contractions. Prolonged use may additionally lead to localised heating and skin burns due to regions of high current density (“hotspots”) underneath the pads, particularly at electrode edges—a phenomenon known as the “edge effect”.

Our project aims to innovatively design external pacing pads that minimise patient discomfort from pain and burns while maintaining effective cardiac capture. We proposed an optimal electrode design incorporating an impedance gradient using a perforated silver/silver chloride anterior electrode. The design consisted of 1500 0.4mm diameter perforations arranged in concentric rings to promote a more uniform current distribution. An anterior-posterior configuration was implemented, with a larger posterior electrode (140 mm diameter) relative to the anterior electrode (100 mm diameter) to improve current delivery across the chest wall.

Computational modelling demonstrated that this design reduced current non-uniformity ($K = 2.76$) and peak current density (peak $|J|$), particularly at electrode edges. The larger posterior electrode further supported effective current transmission ($K = 6.08$ minimum).

Overall, the proposed impedance-gradient electrode design demonstrates a reduction in edge-associated hotspots, indicating a more uniform current distribution and lower risk of thermal injury. These findings suggest that such designs may improve patient comfort and safety during external pacing.

1 Introduction

External pacing is a life-saving intervention used to manage cardiac arrhythmias, including severe bradycardia (heart rate < 40 BPM) and third-degree heart block[1][2]. It is commonly employed as a temporary “bridging” strategy prior to permanent pacemaker implantation[2]. The procedure involves placing two surface electrodes on the chest in either anterior-lateral or anterior-posterior configurations (Figure 1). Electrical current is delivered across the chest wall and gradually increased until cardiac capture is achieved, restoring normal sinus rhythm.[3]

Despite its clinical importance, external pacing is frequently associated with significant patient discomfort.

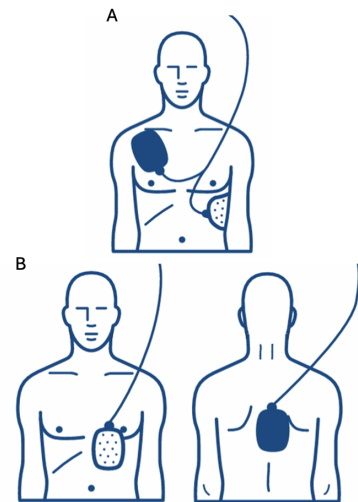


Figure 1: A. Anterio-Lateral Pad Positioning B. Anterior-Posterior Pad Positioning.[3]

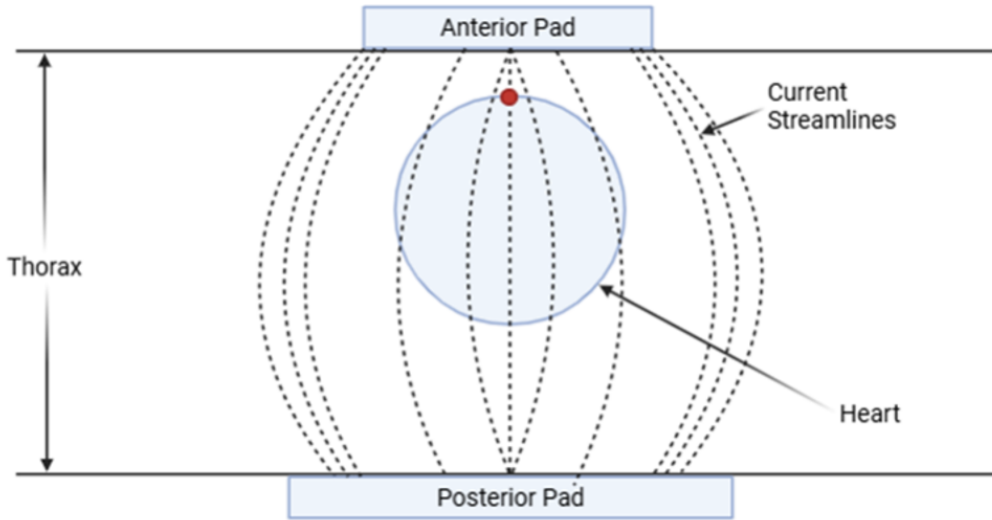


Figure 2: Schematic representing the mechanism behind edge effect burns. The red circle indicates the desired point of capture on the heart. As current is passed across the chest wall, current streamlines converge towards the edges of pads, particularly anteriorly. Image created using <https://www.biorender.com/>

Electrical stimulation can activate skeletal muscle, resulting in painful contractions that often require sedation[1][2]. Prolonged pacing may also cause thermal injury at the electrode-skin interface due to localised regions of high current density – “hotspots”[4]. These are particularly pronounced at electrode edges due to the “edge effect”, where current density increases as current streamlines converge at the electrode perimeter (Figure 2), more commonly observed in anterior pads[5]. This can result in localised heating and skin burns.

Clinical evidence describing pacing-related burns remains limited, with much of the existing literature focused on defibrillation or paediatric populations, and studies investigating underlying mechanisms are often outdated. Nevertheless, case reports highlight the clinical relevance of this issue and identify key contributing factors causing third degree burns, including elevated current density, excessive electrode pressure, inadequate electrolyte gel application, and suboptimal pad design[2]. These factors increase heating at the electrode-skin interface, which is directly proportional to local current density[2]. In sedated patients, the progression of thermal injury may go unnoticed due to the absence of patient-reported pain[2]. Additional factors, including pad positioning, air gaps, and variations in chest wall impedance, may further exacerbate non-uniform current distribution and increase the risk of tissue injury [6]. Collectively, these findings highlight the need to optimise both electrode design and clinical application to minimise patient discomfort.

Previous research has demonstrated that electrode design plays a critical role in determining current

distribution. 3D Finite Element Modelling studies have shown that electrode geometry significantly influences current density patterns, with circular electrodes generally producing more uniform distributions than square designs. Modifications to the electrode-skin interface, including changes in resistivity and thickness, can also alter current flow. Strategies such as incorporating resistive elements or impedance gradients within the electrode have been shown to reduce edge-associated current concentration without compromising overall current delivery.[4]. While originally focused on defibrillation, these findings are directly relevant to external pacing, where minimising current hotspots is essential to reduce pain and thermal injury. More complex electrode designs, such as those incorporating aeration holes, may introduce surface irregularities in current distribution[4]. However, these effects diminish with depth, resulting in relatively uniform current flow in deeper tissues, including the myocardium[4]. Further studies have demonstrated that increasing resistance at the electrode perimeter can redistribute current more evenly across the electrode surface, via resistive coatings or impedance gradients[5] to mitigate the “edge effect”. While such approaches improve current uniformity, they must be carefully balanced against maintaining sufficient current to achieve effective cardiac capture.

Overall, there remains a need for optimised external pacing electrode designs that reduce current non-uniformity while maintaining effective cardiac stimulation. The multi-material composition of electrode pads further complicates this challenge, requiring careful consideration of electrical, mechanical, and geometrical properties.

Therefore, our project aims to design and evaluate novel external pacing electrodes that minimise patient discomfort by reducing edge-associated current “hotspots” and improving current distribution across the electrode surface. To achieve this, we will be using an impedance gradient design with a focus on the following specifics:

- Mitigate the “Edge Effect” using a perforated Ag/AgCl conductive layer on a Mylar substrate creating a resistive gradient to redistribute current density away from electrode edges.
- Computational finite element modelling, coupled with multi-objective optimisation (NSGA-II), to determine optimal perforation geometry for balancing current uniformity and peak tissue current density.
- Refine pad configuration by evaluating the anterior-posterior arrangement with asymmetrical sizing to further reduce peak current density.
- A materials evaluation using conductive hydrogels (such as PEDOT:PSS) are investigated to enhance electrode–skin coupling and promotion of uniform current distribution.
- Experimental validation undertaken with a gelatin-saline tissue phantom to quantitatively assess thermal performance relative to standard electrodes using temperature sensitive ink.

We hypothesize that a gradient impedance perforated design will significantly reduce current non-uniformity ($K = J_{max}/J_{min}$) and peripheral hotspots compared to standard electrodes, thereby decreasing pain and thermal injury.

2 Methods

2.1 Perforation Design and Conductive Layer Fabrication

The pad was designed to reduce current concentration at the electrode periphery by introducing a gradient impedance within the conductive layer. Previous studies have shown that increased peripheral impedance can reduce edge effects and lower peak temperature during transcutaneous stimulation [4][7]. The final design comprised a 10 cm circular front pad with a perforated conductive metallic layer and a 14 cm circular back pad intended to spread current over a larger area and reduce peak current density.

The perforation pattern (Figure 3) consisted of eight concentric rings of holes, with the number of holes increasing by 25 in each successive ring, starting from 100

holes in the innermost ring. This gave a total of 1500 holes, each with a diameter of 0.4 mm. The diameters of the innermost and outermost rings were 6.0 cm and 9.5 cm, respectively, with the diameter of each successive ring increasing by 5 mm. This arrangement was intended to reduce conductive area towards the pad periphery and therefore increase edge impedance.

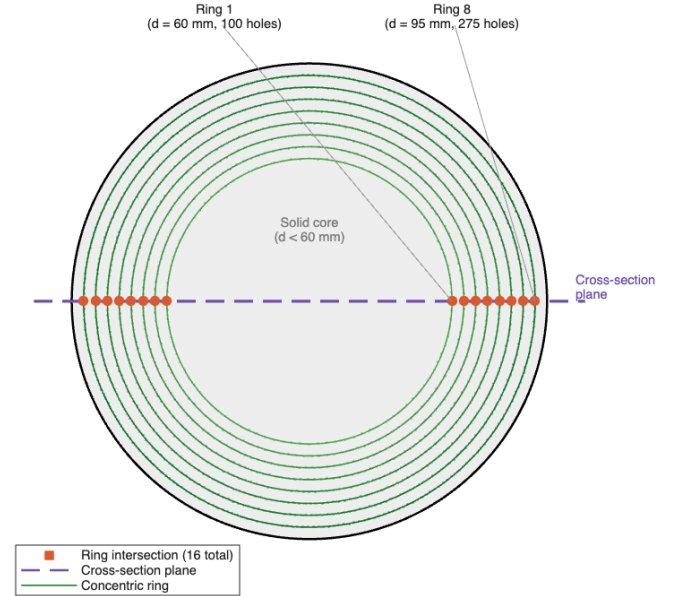


Figure 3: Schematic diagram of the concentric-ring perforation pattern of the Ag/AgCl conductive layer.

Mylar film with a thickness of 0.125 mm (RS Components, 7850798) was selected as the carrier substrate for the conductive layer. The 2D perforation sketch was exported from Fusion 360 as a DXF file, and a Universal Laser Systems VLS2.30 CO₂ laser cutter/engraver was used to cut and perforate the Mylar film. This laser-cut substrate formed the basis of the conductive metallic layer.

A fabrication protocol was then established for deposition of the Ag/AgCl coating onto the laser-cut Mylar. In the planned method, Ag/AgCl paste with a 60:40 ratio (Sigma-Aldrich, 901773) would be manually applied to the laser-cut Mylar using a plastic palette knife, before drying at 80°C for 10 minutes to evaporate the solvent and leave a conductive layer adhered to the substrate. However, due to delays in delivery of the Ag/AgCl paste, this coating stage was not completed before the report deadline. As a result, only the perforation design and laser-cut Mylar preparation were completed for this submission.

2.2 Pad Layer Configuration

The pacing pad comprised a multilayer electrode assembly, with each component identified in Figure 4. The insulating plate (1) was made from high-density

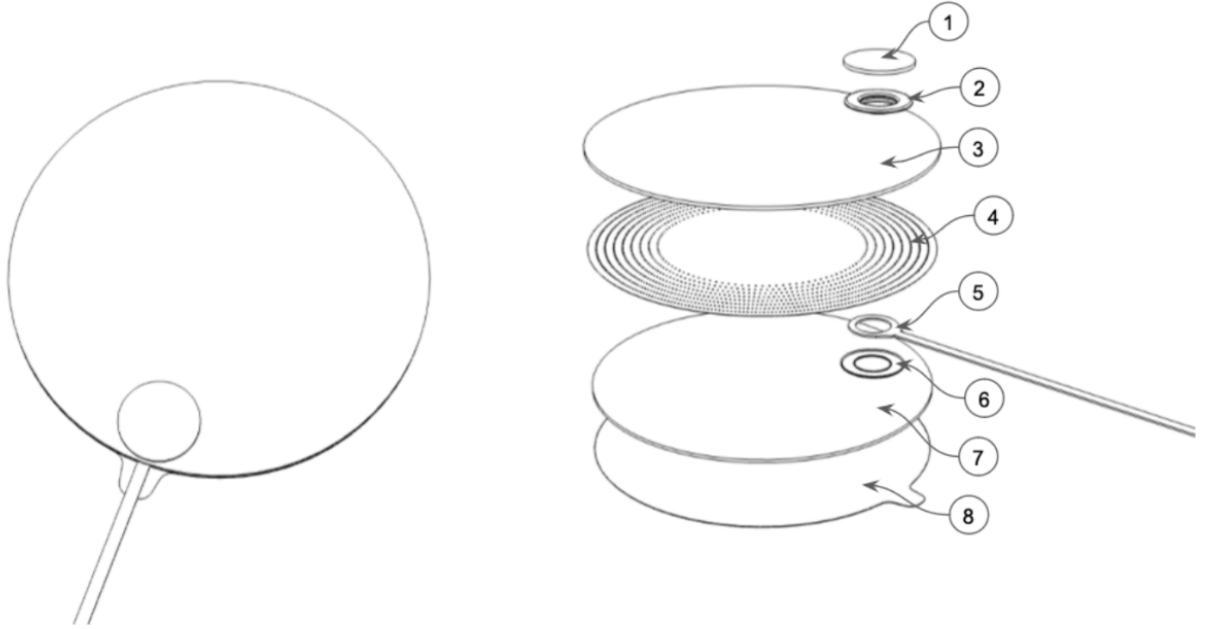


Figure 4: Exploded diagram of the transcutaneous pacing pad design (1) Insulating plate (2) Eyelet (3) Backing (4) Ag/AgCl coated Mylar film (5) Ring terminal wire (6) Washer (7) Hydrogel (8) Release film

white foam and provided a non-conductive structural support layer. Electrical connection to the pad was achieved using a brass eyelet (2) and a brass/copper ring terminal wire (5), with a brass washer (6) used to secure and reinforce the terminal connection. The backing (3) provided mechanical support to the overall pad structure. The skin-contacting layer consisted of a PEDOT:PSS composite hydrogel (7), which formed the electrode-skin interface. A release film (8) covered the hydrogel before use and was removed immediately prior to application.

2.3 Computational Domain and Governing Equation

$$-\nabla \cdot [\sigma(x, z) \nabla V] = 0 \quad (1)$$

The current density distribution was simulated using a two-dimensional finite-element model of the human thorax, following the methodology of Krasteva and Papazov [4]. A 2D sagittal cross-section was adopted because the edge effect is principally a radial phenomenon: current density varies most strongly along the electrode width rather than around its circumference, making a cross-sectional slice sufficient to capture the key redistribution behaviour while keeping computational cost low enough for iterative optimisation. The computational domain was a rectangular cross-section ($0.50 \text{ m} \times 0.25 \text{ m}$) representing a simplified sagittal thorax slice, with a circular region of radius 45 mm at a depth of 95 mm representing cardiac tissue. Two Ag/AgCl gel electrode pads, each 100 mm wide and 1 mm thick, were positioned centrally on the upper and lower boundaries.

The steady-state potential $V(x, z)$ was governed by

the generalised Laplace equation (Eq 1), where σ is the spatially varying conductivity. Bulk tissue was assigned $\sigma = 0.2 \text{ S/m}$ and cardiac muscle 0.5 S/m , consistent with published thoracic conductivity ranges for low-frequency stimulation [4]. The electrode metal layer was assigned $\sigma = 0.2 \text{ S/m}$, derived from the Ag/AgCl surface resistivity $\rho = 5 \Omega$ and layer thickness $d = 1 \text{ mm}$. The tissue was modelled as a homogeneous bulk conductor; while this does not capture the layered structure of skin, fat, muscle, and bone, it is appropriate for comparative design evaluation, where the relative difference between electrode configurations matters more than absolute current density values. Dirichlet boundary conditions were fixed at $V = 0 \text{ V}$ on the ground electrode and $V = 10 \text{ V}$ on the active electrode, with homogeneous Neumann conditions on all remaining boundaries. The magnitude of current density, $|J|$, was computed by Eq 2, using the MATLAB Partial Differential Equation Toolbox.

$$|J| = |-\sigma \nabla V| \quad (2)$$

2.4 Perforation Model

Electrode perforations were modelled as discrete insulating regions ($\sigma = 10^{-8} \text{ S/m}$) within the metal layer. This was implemented by modifying the conductivity coefficient function at each mesh evaluation point: any point falling within a perforation was assigned the insulating conductivity, while all other electrode points retained the nominal Ag/AgCl conductivity of 0.2 S/m .

2.5 Performance Metrics

Electrode performance was assessed using two quantitative metrics. The first was the non-uniformity ratio (Eq 3) and the second was the peak current density.

$$K = \frac{J_{\max}}{J_{\min}} \quad (3)$$

The non-uniformity ratio was evaluated along a horizontal line 0.5 mm below the electrode-tissue interface across the full electrode width. K quantifies the evenness of current injection into tissue; $K = 1$ corresponds to perfectly uniform delivery, and higher values indicate greater non-uniformity. The second metric was the peak current density in tissue, $\max |J|$, computed across all bulk tissue mesh points (excluding the cardiac region and electrode metal layers). Both metrics were compared to a solid (unperforated) electrode baseline to assess the effect of perforation geometry.

2.6 Multi-Objective Optimisation of Perforation Geometry

To identify the perforation geometry that best reduces edge-current concentration whilst limiting peak current density, three design variables were optimised simultaneously using the Non-dominated Sorting Genetic Algorithm II (NSGA-II) [8]. The design variables were:

- **Hole radius** (0.2–2.5 mm): the radius of each circular perforation in the electrode metal layer.
- **Number of holes per side** (2–30, integer-constrained): the number of perforations distributed between r_{perf} and the electrode edge on each side of the electrode centre.
- **Perforation start radius**, r_{perf} (5–45 mm): the radius of the central solid (unperforated) region, measured from the electrode centre. Perforations begin at this radius and extend to the electrode edge.

Holes were uniformly distributed between r_{perf} and the electrode edge, with each hole radius limited to 40% of the inter-hole spacing to prevent overlap. The two objectives—minimisation of K and minimisation of peak tissue $|J|$ —were simultaneously optimised, yielding a Pareto front of non-dominated solutions representing the best achievable trade-offs between uniformity and peak current density.

The optimisation used a population of 60 over 40 generations, with crossover fraction 0.80 and Pareto fraction 0.35, giving approximately 2,400 model evaluations. A coarse mesh ($H_{\max} = 2.0$ mm, $H_{\min} = 0.5$ mm)

was used during optimisation, and selected Pareto-optimal designs were re-solved on a finer mesh ($H_{\max} = 1.2$ mm, $H_{\min} = 0.3$ mm). Three representative designs were retained for comparison: the minimum- K solution, the minimum-peak- $|J|$ solution, and a compromise knee-point solution identified as the Pareto member closest to the origin in min-max normalised objective space.

2.7 Concentric-Ring Electrode Design

A manufacturable perforated electrode geometry was then derived from the optimisation outputs using 8 concentric rings of circular holes. Ring diameters ranged from 60 to 95 mm in 5 mm spaces between them. The hole counts increased from 100 (innermost) to 275 (outermost) in increments of 25, giving 1,500 perforations per electrode (Table 1). The central region (<60 mm diameter) remained unperforated.

Ring	Diameter (cm)	Radius (mm)	Holes per ring	Spacing (mm)
1	6.0	30.0	100	—
2	6.5	32.5	125	5.0
3	7.0	35.0	150	5.0
4	7.5	37.5	175	5.0
5	8.0	40.0	200	5.0
6	8.5	42.5	225	5.0
7	9.0	45.0	250	5.0
8	9.5	47.5	275	5.0

Table 1: Concentric ring perforation specification. The central region (diameter < 60 mm) remains solid.

2.8 Ring Electrode Simulation and Hole Diameter Sweep

Figure 3 shows the cross-section plane intersected each of the eight concentric rings in the electrode at two points (one on either side of the electrode centre) giving 16 intersection points in total. At each intersection point, a perforation of the corresponding physical hole diameter was modelled as a discrete insulating region. Simulations were repeated for a range of hole diameters with all cases solved using the fine mesh.

2.9 Back Electrode Length Sweep

To assess the effect of posterior electrode size, the back (bottom) electrode width was varied from 60 to 240 mm

in increments of 20 mm, while the front (top) electrode width was fixed at 100 mm. All other geometric, tissue, and mesh parameters were held constant.

The concentric-ring perforation pattern was represented using a spatially varying surface impedance rather than explicitly meshing individual holes. Each electrode was assigned a central electrode impedance of $0.02 \Omega \cdot \text{m}^2$ and a peripheral impedance of $0.20 \Omega \cdot \text{m}^2$, with a smooth transition of half-width $\delta = 0.5 \text{ mm}$ located 4 mm inward from the pad edge. This continuous impedance profile approximated the current-limiting effect of densely perforated outer rings identified by the NSGA-II optimisation, while avoiding the computational cost of resolving each perforation. Dirichlet boundary conditions of 0 V at the back electrode and 5 V at the front electrode were applied.

Performance metrics were extracted from the post-processed current density field. This included an edge hotspot metric which defined current density at the electrode edges as the 95th percentile of $|J|$ within a 3 mm strip at each pad edge near the skin surface. In addition to this, the mean current density at the heart wall (calculated over a 5 mm annular shell surrounding the cardiac inclusion), and the non-uniformity coefficient, K , were also extracted.

2.10 Pad Testing Protocol

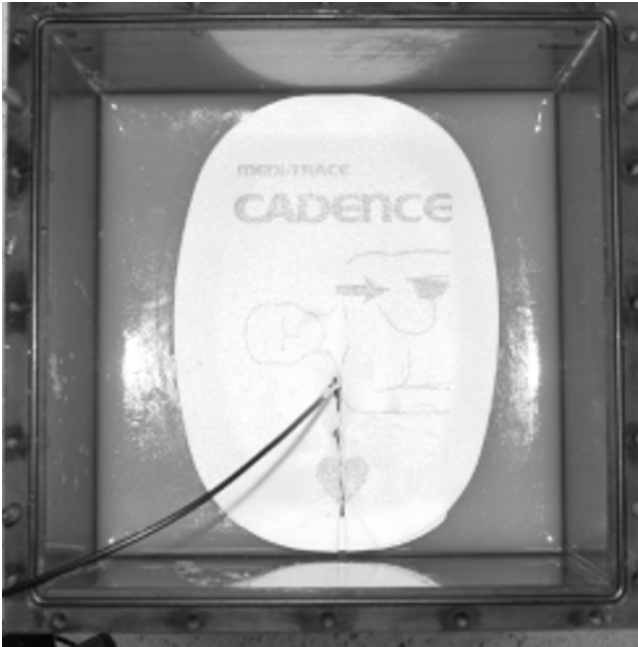


Figure 5: Cube-shaped phantom partially filled with gel. Electrode is placed on the surface of the gel with wire attached to center.[9]

The thermal performance and edge-effect mitigation of the perforated pacing pad will be evaluated using a gelatin-saline phantom model integrated with thermosensitive visualisation. This approach is selected to enable spatial mapping of heat distribution at the electrode-tissue interface. However, due to delays in delivery of the Ag/AgCl paste and hydrogel, the full pad prototype could not be completed before the report deadline, so this testing was not carried out within the timeframe of this submission. Fabrication and protocol follow the procedures performed by DeMonte[9].

2.10.1 Phantom fabrication

The phantom setup will consist of a conductive copper plate placed at the base of the phantom container to act as the return electrode. The tissue-mimicking phantom will be prepared by mixing 2250 mL of deionised water, 450 g of animal-hide gelatin, 20.25 g of NaCl, and 30 mL of Dettol. The mixture will be heated and stirred until the gelatin and NaCl are fully dissolved, then poured to a height of approximately 9 cm.

Once the phantom has set, a thin and uniform layer of temperature-sensitive ink will be applied to the surface. This ink will act as the thermal indicator, remaining opaque at room temperature and becoming gradually transparent when the temperature exceeds 30°C . This will provide a visual map of hotspot formation across the pad-phantom interface.

2.10.2 Experimental setup and instrumentation

To capture the thermal profile across the surface, a digital camera will be positioned to record the interface through the pad during stimulation. This will provide a continuous visual map of heat generation rather than relying on measurements at a small number of discrete points. A series shunt resistor will also be included in the circuit to allow real-time monitoring of the delivered current.

2.10.3 Measurement protocol

Testing will be carried out only once the phantom has reached room temperature, so that the ink remains below its 30°C activation threshold. The test pad will then be attached to the inked phantom surface using controlled pressure to maintain repeatable contact conditions. The pacing pad will be connected to a programmable current pulse generator, while the copper plate at the base of the phantom will be connected to the return terminal.

The testing sequence will consist of three stages.

First, a low-energy pulse train will be applied as a preliminary check to confirm stable contact and correct current delivery. This will be followed by formal stimulation using 14 ms pulses at 200 mA, consistent with established edge-effect imaging conditions. During stimulation, digital video will be recorded to capture the progression and location of transparency in the temperature-sensitive ink. Simultaneously, the voltage across the pad-phantom system and the current through the shunt resistor will be recorded.

2.10.4 Data analysis and performance metrics

The effectiveness of the perforated design in reducing edge-related heating will be assessed using several metrics. Hotspot distribution will be evaluated from the spatial pattern of ink transparency, with particular attention to whether the 30°C threshold is reached first at the pad edge or at the centre. Effective impedance will be calculated from the recorded voltage and current during stimulation. Impedance drift over repeated pulsing will also be monitored to assess the stability of the electrode-phantom interface.

Each pad configuration will be tested at least three times in a randomised order. A constant rest period will be maintained between trials to allow the phantom to return to a consistent thermal baseline and for the ink to recover to its opaque state. This protocol will be used in future work once the full prototype has been assembled.

Statistical comparison will be performed between the perforated prototype and the standard Mindray control pads to assess whether the redesigned pad produces a lower temperature rise at the electrode edge during the 200 mA stimulation protocol. Given the planned sample size of $n = 3$ per condition and the unknown variance of the phantom's thermal response, a two-sample Student's t -test would be used. The null hypothesis will be that there is no significant difference in mean edge temperature rise between the perforated prototype and the solid control pad, while the alternative hypothesis will be that the perforated design produces a lower mean edge temperature rise due to redistribution of current density. A significance level of $\alpha = 0.05$ will be used. To assess repeatability, the coefficient of variation will also be calculated for each trial set to monitor possible experimental drift caused by phantom dehydration or inconsistent pad-to-gel contact pressure.

3 Results

3.1 Baseline Simulation and Design Rationale

Figure 6 shows the simulated current density field beneath the front electrode for both the standard solid electrode and the perforated prototype design. With the solid electrode, pronounced hotspots are visible at both edges, consistent with the edge effect in planar conductors: current streamlines converge at the electrode perimeter, producing localised regions of high current density that increase the risk of thermal injury and patient pain. The perforated design reduces hotspot intensity at both edges, producing a more uniform distribution across the electrode width. Small regions of locally reduced current density at the perforation locations confirm that the insulating holes redirect current flow as intended.

These baseline results motivated the identification of three geometric parameters that control the redistribution effect (illustrated in Figure 3): (i) the *hole radius*, which determines how much conductive area is removed at each perforation; (ii) the *number of holes per side*, which sets how densely the perforations are distributed between the solid centre and the electrode edge; and (iii) the *perforation start radius* r_{perf} , the radial distance from the electrode centre at which perforations begin, where a smaller r_{perf} means perforations extend further towards the centre. These three variables were carried forward into the multi-objective optimisation described in Section 3.2.

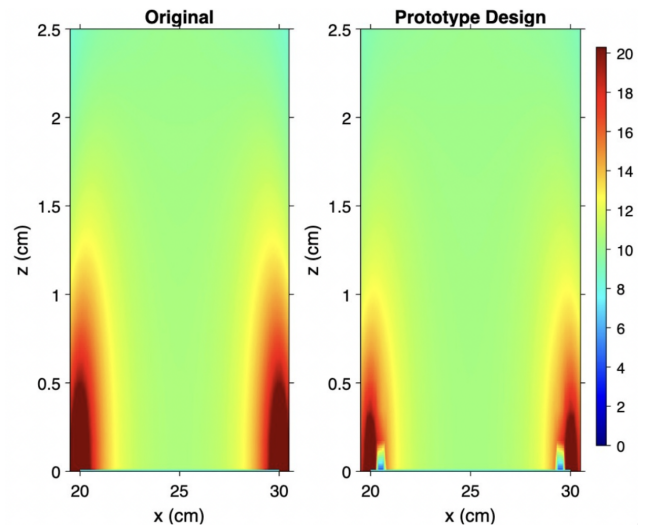


Figure 6: Simulated current density magnitude $|J|$ (A/m^2) in tissue beneath the front electrode for (left) a standard solid electrode and (right) the perforated prototype design. Colour scale: 0–20 A/m^2 .

3.2 Multi-Objective Optimisation of Perforation Geometry

Using the three design variables identified in Section 3.1, a multi-objective optimisation was performed to find the perforation geometry that best balances two competing goals: reducing current non-uniformity (quantified by $K = J_{\max}/J_{\min}$, where $K = 1$ represents perfectly uniform delivery) and minimising the peak current density in tissue (peak $|J|$), which is directly related to localised heating and burn risk. The three variables, namely hole radius, number of holes per side, and perforation start radius r_{perf} (see Figure 3), were optimised simultaneously using the NSGA-II genetic algorithm [8] across approximately 2,400 candidate designs over 40 generations (Figure 7a). The solid (unperforated) baseline yielded $K = 2.56$ and peak $|J| = 35.14 \text{ A/m}^2$. No perforated configuration simultaneously improved both objectives; the solid design sat on the Pareto front itself. The front spanned $K \approx 2.33$ – 2.60 and peak $|J| \approx 34.6$ – 35.4 A/m^2 . Pareto-optimal solutions clustered at hole radii of 1.5 – 2.5 mm and 0.5 – 1.0 mm , low r_{perf} ($< 10 \text{ mm}$), and 15 – 25 holes per side (Figure 7b), though two solutions had $r_{\text{perf}} \approx 45 \text{ mm}$.

Three representative solutions from the Pareto front are summarised in Table 2. The best-uniformity design ($K = 2.33$, -9.0%) used large holes (2.12 mm radius), 21 per side, and $r_{\text{perf}} = 5.5 \text{ mm}$, but peak $|J|$ rose 0.9% to 35.44 A/m^2 . The knee-point ($K = 2.40$, -6.3%) achieved similar improvement with peak $|J|$ essentially unchanged, using 2.00 mm holes at $r_{\text{perf}} = 5.7 \text{ mm}$. The best peak $|J|$ design used smaller holes (0.77 mm radius) confined to a narrow outer ring ($r_{\text{perf}} = 44.2 \text{ mm}$), reducing peak $|J|$ by 1.5% but with K slightly worse than solid. In practical terms, the 9% reduction in K achieved by the best-uniformity design corresponds to a measurable redistribution of current away from the electrode edges, which would be expected to lower localised heating and reduce the risk of skin burns during prolonged pacing.

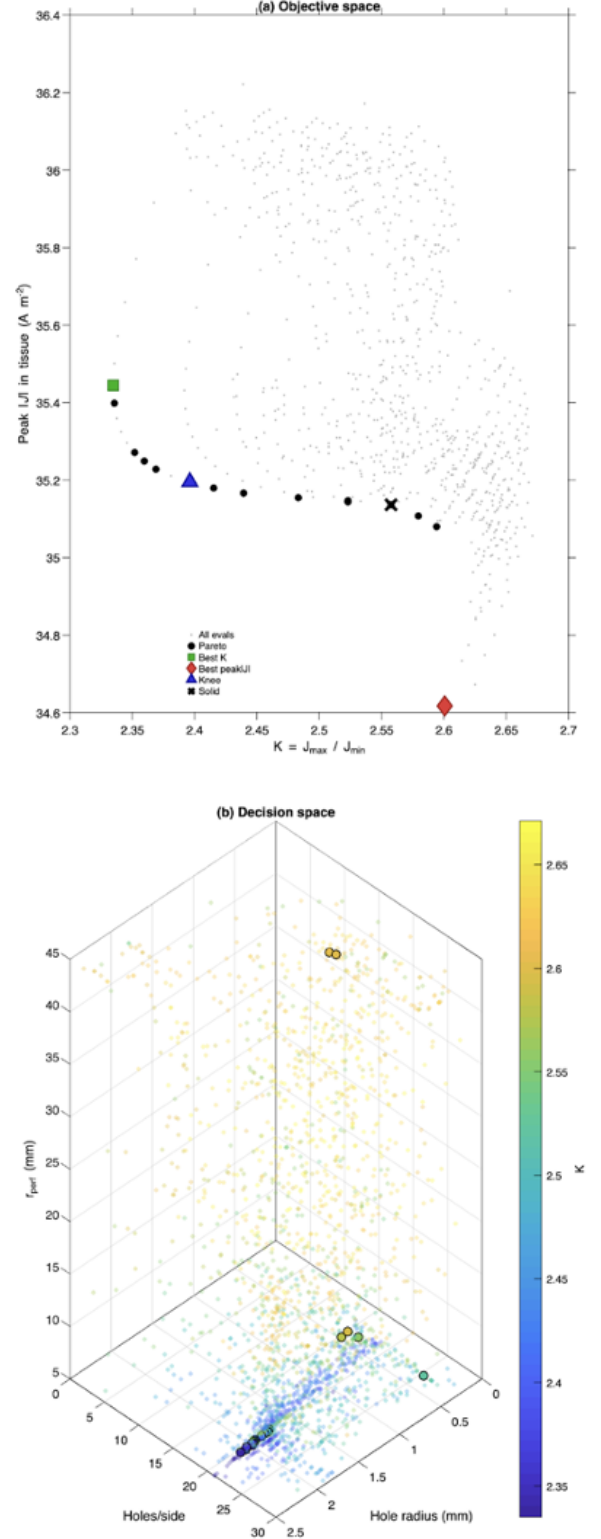


Figure 7: NSGA-II optimisation results. (a) Objective space showing all designs (grey) and the Pareto front (black). The solid baseline (cross) sits at $K = 2.56$, peak $|J| = 35.14 \text{ A/m}^2$. The best- K solution ($K = 2.33$), knee point ($K = 2.40$), and best peak $|J|$ solution (peak $|J| = 34.62 \text{ A/m}^2$) are highlighted. (b) Decision space coloured by K , showing Pareto-optimal solutions clustering at high hole radius (1.5 – 2.5 mm), low r_{perf} ($< 10 \text{ mm}$), and moderate hole count (15 – 25 per side).

Design	Hole radius (mm)	Holes/side	r_{perf} (mm)	K	Peak $ J $ (A/m ²)	ΔK (%)
<i>Solid baseline</i>	—	0	—	2.56	35.14	—
Best uniformity	2.12	21	5.5	2.33	35.44	−9.0
Knee point	2.00	21	5.7	2.40	35.19	−6.3
Best peak $ J $	0.77	18	44.2	2.60	34.62	+1.6

Table 2: Key Pareto-optimal solutions from the NSGA-II optimisation. ΔK is relative to the solid baseline; negative values indicate improved uniformity.

3.3 Parameter Sensitivities

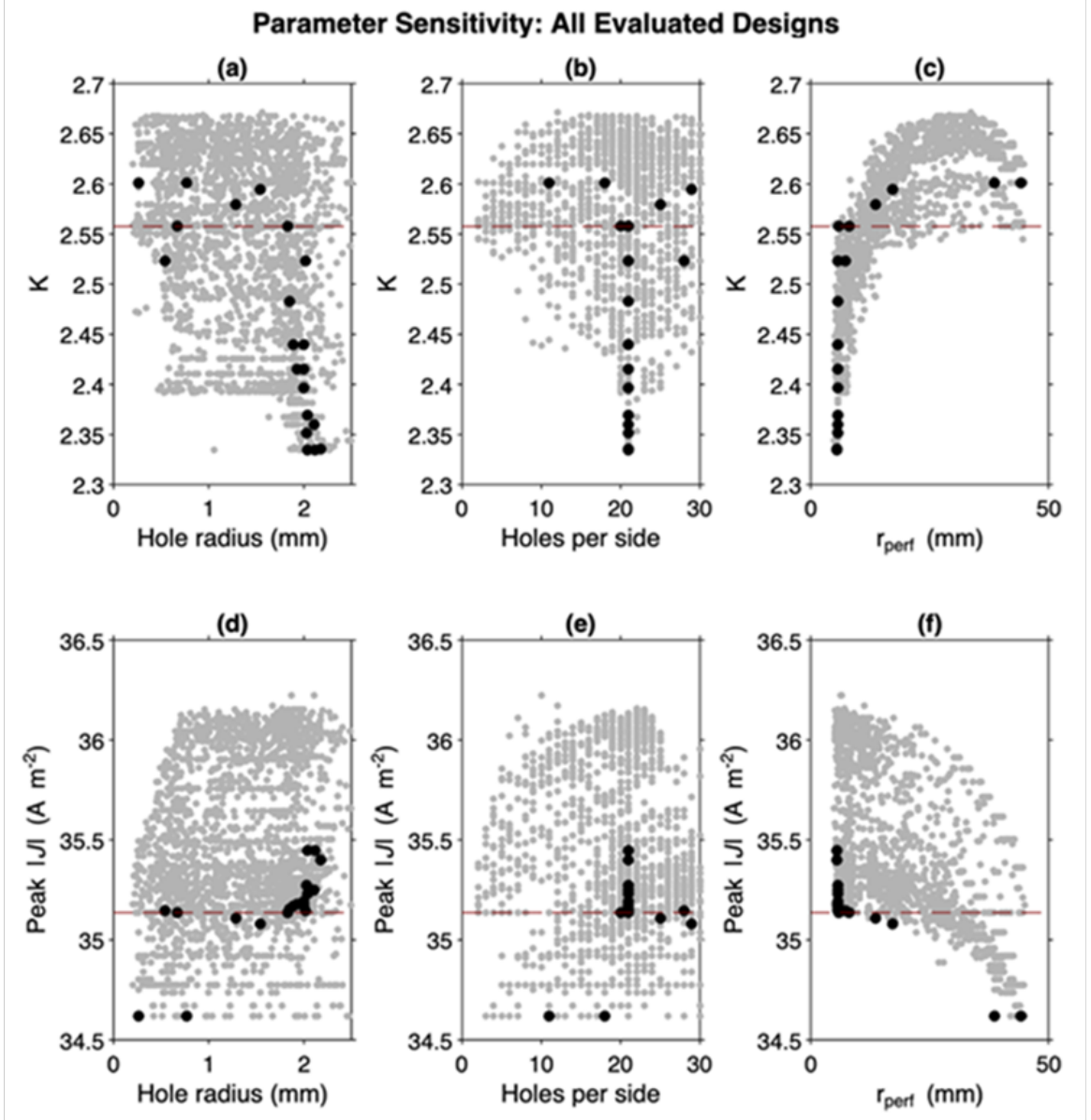


Figure 8: Parameter sensitivity across all 2,400 evaluated designs. Grey dots: all valid evaluations; black dots: Pareto-optimal solutions; dashed red line: solid baseline. Top row: non-uniformity ratio K versus (a) hole radius, (b) holes per side, and (c) perforation start radius r_{perf} . Bottom row: peak tissue current density versus (d) hole radius, (e) holes per side, and (f) r_{perf} . Perforation start radius (c, f) has the strongest single-variable influence on K , with all Pareto-optimal solutions below $r_{\text{perf}} = 10$ mm. Hole radius (a) shows a nonlinear threshold, with K declining steeply above approximately 1.5 mm. Large-radius designs (d) exhibit a wide spread in peak $|J|$, reflecting the trade-off between uniformity and current crowding at hole edges.

The sensitivity of both objectives to each design variable was assessed across all 2,400 evaluated designs (Figure 8).

Perforation start radius r_{perf} was the most influential single variable (Figure 8c,f). Designs with $r_{\text{perf}} < 10$ mm—meaning perforations extending close to the electrode centre—achieved K as low as 2.30, while those with $r_{\text{perf}} > 30$ mm could not improve upon the solid baseline. This confirms that the redistribution effect requires perforations across a substantial fraction of the electrode area, not just at the periphery.

Hole radius exhibited a nonlinear threshold (Figure 8a). K remained near the baseline for holes smaller than ~ 1.5 mm but declined steeply above this size. However, large-radius designs also showed elevated peak $|J|$ due to current crowding at hole edges (Figure 8d), which is the physical origin of the uniformity–peak current density trade-off seen in the Pareto front.

Hole count had the weakest influence (Figure 8b,e); the optimiser converged on 18–21 per side, consistent with the dominance of a small number of geometric parameters reported in global sensitivity analyses of related bioelectromagnetic models [10].

3.4 Concentric-Ring Electrode: Effect of Hole Diameter

The NSGA-II optimisation identified favourable perforation parameters but used a simplified linear arrangement. To translate these findings into a manufacturable geometry, a concentric-ring layout was developed (Figure 3, Table 1), in which eight rings of perforations are arranged around a solid central core, with hole density increasing towards the electrode periphery to concentrate impedance where the edge effect is strongest. This design (Figure 3, Table 1), comprising eight rings of perforations with hole counts increasing from 100 (innermost) to 275 (outermost) to concentrate impedance at the periphery, was simulated across a range of hole diameters to determine how perforation size affects current uniformity (K) and peak tissue current density in the manufacturable layout. The solid baseline for this geometry yielded $K = 2.82$ and peak $|J| = 36.84$ A/m² (Table 3); the difference from the NSGA-II baseline ($K = 2.56$) arises from the finer mesh resolution used in this simulation. The diameter sweep revealed a non-monotonic response (Table 3). At 0.2–0.3 mm, K decreased by 2.5–2.6% with negligible peak $|J|$ change. At the selected manufacturing baseline of 0.4 mm (matching the CO₂ laser cutter’s minimum feature size), K was 2.76 ($\Delta K = -2.3\%$). Between 0.6 and 1.0 mm, the improvement diminished as current crowding offset the redistribution benefit. At 1.5 mm, K dropped to its minimum of 2.66 (−5.8%).

At 2.0 mm, performance degraded sharply: K rose to 3.30 (+16.9%) and peak $|J|$ to 38.74 A/m² (+5.1%). Each hole spanned 2.0 mm of the 2.5 mm inter-ring gap, leaving only 0.5 mm of solid metal between perforations. The resulting near-continuous insulating barrier forced current to concentrate at the narrow inter-hole gaps and the pad edge.

d_{hole} (mm)	K	Peak $ J $ (A/m ²)	ΔK (%)	$\Delta \text{peak } J $ (%)
<i>Solid</i>	2.82	36.84	—	—
0.1	2.82	36.85	< 0.1	< 0.1
0.2	2.75	36.79	−2.5	−0.1
0.3	2.75	36.80	−2.6	−0.1
0.4	2.76	36.88	−2.3	+0.1
0.6	2.77	36.95	−2.0	+0.3
0.8	2.78	37.08	−1.5	+0.6
1.0	2.80	37.25	−0.9	+1.1
1.5	2.66	37.31	−5.8	+1.3
2.0	3.30	38.74	+16.9	+5.1

Table 3: Hole diameter sweep results for the concentric-ring design. Negative Δ values indicate improvement relative to solid.

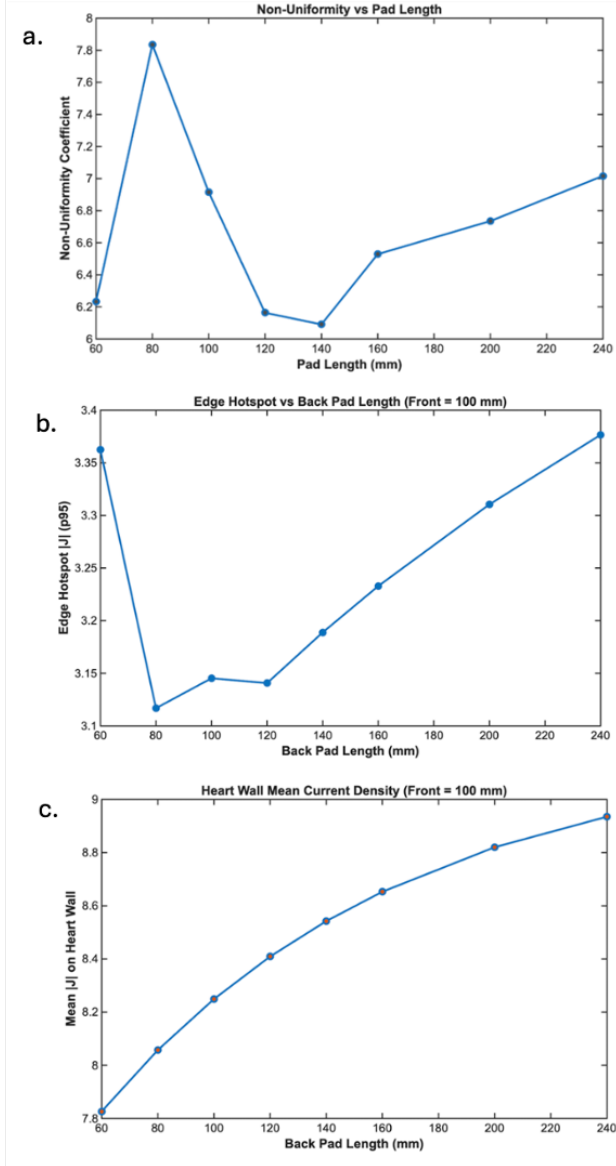


Figure 9: (a) Non-uniformity coefficient K as a function of back pad length, with the front pad held at 100 mm. K peaks at 80 mm, reaches a minimum of 6.08 at 140 mm, and rises again at larger pad widths. (b) Edge hotspot $|J|$ (95th percentile) as a function back pad length, with front pad fixed at 100 mm. It reaches a minimum of 3.12 A/m^2 at 80 mm, plateaus to 120 mm, and then increases steadily. (c) Heart-wall mean $|J|$ as a function of back pad length, with the front pad held at 100 mm. Cardiac current density increases monotonically from 7.82 A/m^2 at 60 mm to 8.94 A/m^2 at 240 mm, with a saturating profile.

3.5 Back Electrode Length Sweep

In the anterior-posterior pad configuration (Figure ??), the posterior (back) electrode is placed behind the patient and can be made larger than the anterior pad without practical constraint. A larger back electrode

spreads return current over a wider area, potentially reducing peak current density at the front pad where edge-effect burns are most likely. The effect of posterior electrode size was therefore assessed using the impedance-modulated electrode model, with the front electrode fixed at 100 mm and the back electrode swept from 60 to 240 mm.

The non-uniformity coefficient exhibited a non-monotonic dependence on back pad length (Figure 9a). K increased from 6.22 at 60 mm to a peak of 7.83 at 80 mm, then decreased to a minimum of 6.08 at 140 mm before rising again to 7.02 at 240 mm. The elevated K at 80 mm suggests that the high-impedance annular zone occupies a disproportionate fraction of the electrode area at this width, constraining current flow to a narrow central region. The minimum at 140 mm indicates a more favourable balance between central conduction area and peripheral current.

The edge hotspot exhibited a shallow U-shaped profile (Figure 9b), decreasing from 3.36 A/m^2 at 60 mm to a minimum of 3.12 A/m^2 at 80 mm before rising steadily to 3.38 A/m^2 at 240 mm. The initial decrease reflects the back pad edges moving away from the high-flux central axis. The subsequent rise at larger widths is consistent with the electrode edges extending into regions where domain boundaries confine current flow, compressing flux lines near the periphery.

The heart-wall mean $|J|$ increased monotonically from 7.82 A/m^2 at 60 mm to 8.94 A/m^2 at 240 mm with a saturating profile (Figure 9c), reflecting the larger current capture area of wider electrodes. The rate of increase diminished above 160 mm, consistent with the domain width imposing a geometric ceiling on additional current gathered by further pad expansion.

4 Discussion

This project successfully demonstrated, through computational modelling, that a perforated impedance-gradient electrode design reduces edge-associated current density hotspots compared with standard solid electrodes. Using finite-element simulation coupled with NSGA-II multi-objective optimisation, we identified the key geometric parameters governing current redistribution and quantified the trade-offs between uniformity and peak tissue current density across approximately 2,400 candidate designs. These results were translated into a manufacturable concentric-ring electrode (Figure 3) with verified performance across a range of hole diameters and posterior electrode sizes. A complete manufacturing workflow, including laser-cut Mylar substrate preparation, has been established and partially executed.

4.1 Optimisation to Manufacturable Design

The NSGA-II optimisation [8, 11] identified three principal design levers. Perforation start radius (r_{perf} , defined in Figure 3) was the most influential: every Pareto-optimal solution had $r_{\text{perf}} < 10$ mm, and increasing it above 30 mm gave no improvement over a solid electrode. Hole radius had a nonlinear effect, with the non-uniformity ratio K dropping sharply only once holes exceeded 1.5 mm radius, so optimal solutions clustered around 2.0–2.5 mm. Hole count was the least sensitive parameter, settling at 18–21 per side across most solutions.

These findings directly shaped the concentric-ring electrode layout (Figure 3). Perforation density increases from 100 holes (innermost ring, 60 mm diameter) to 275 (outermost, 95 mm), concentrating current redistribution at the periphery where the edge effect is most pronounced. The innermost ring at 30 mm radius is larger than the optimisation preferred, a deliberate compromise to ensure consistent Ag/AgCl paste coating and hydrogel contact across the solid central region. The 0.4 mm hole diameter matches the CO₂ laser cutter’s minimum feature size, with the diameter sweep (Table 3) confirming $\Delta K = -2.3\%$ and robustness to ± 0.1 mm variation.

4.2 Current Density Redistribution

The baseline simulation (Figure 6) confirmed that a solid electrode produces pronounced current-density hotspots at the electrode edges, consistent with analytical predictions of the edge effect in planar conductors. The perforated prototype design visibly reduced hotspot intensity at both edges, with the current density field becoming more uniform beneath the electrode. This represents a key result: the impedance-gradient concept successfully redirects current away from the electrode periphery, directly addressing the mechanism responsible for edge-effect burns. The qualitative improvement was quantified through the non-uniformity ratio K and the peak current density metrics, both of which confirmed the visual impression of improved uniformity.

4.3 Trade-Off Between Uniformity and Peak Current Density

A key finding of the optimisation was the identification and quantification of a fundamental trade-off between current uniformity and peak tissue current density. No perforated design improved both K and peak $|J|$ simultaneously, but the optimisation systematically mapped the achievable design space. The best- K re-

sult ($K = 2.33$, $\Delta K = -9.0\%$) used large holes of 2.12 mm radius starting at $r_{\text{perf}} = 5.5$ mm, but peak $|J|$ increased by 0.9%. Conversely, a design with smaller holes (0.77 mm) confined to a narrow outer ring reduced peak $|J|$ by 1.5% while uniformity worsened. The knee-point design ($K = 2.40$, $\Delta K = -6.3\%$) lay between these extremes, maintaining peak $|J|$ nearly unchanged while still delivering a meaningful improvement in uniformity. The underlying mechanism is localised current crowding at hole edges: insulating perforations redirect current flow but concentrate it at the boundaries of each hole. This trade-off is consistent with finite-element studies linking electrode geometry, current density, and discomfort in transcutaneous stimulation [12], and the Pareto front generated here provides a quantitative basis for selecting the most appropriate design point for a given clinical priority.

4.4 Trade-Off Between Uniformity and Back Electrode Length

Varying back electrode length introduced a separate set of trade-offs. With the front pad fixed at 100 mm, K reached a minimum of 6.08 at 140 mm before increasing to 7.02 at 240 mm. Heart-wall mean $|J|$ increased steadily across the full range, from 7.82 to 8.94 A/m², while the edge hotspot decreased to a minimum of 3.12 A/m² at 80 mm before rising again at larger widths. No single length optimised all three metrics simultaneously.

4.5 Diameter Dependence and Design Limitations

The diameter sweep (Table 3) revealed three distinct regimes. Below 0.6 mm, larger holes improved K by up to 2.6% with negligible change in peak $|J|$. Between 0.8 and 1.0 mm, current crowding began to offset the redistribution benefit. At 1.5 mm, an anomalous improvement appeared ($\Delta K = -5.8\%$), likely due to a resonance between hole width and ring spacing. At 2.0 mm, performance degraded sharply ($\Delta K = +16.9\%$, $\Delta_{\text{peak}} |J| = +5.1\%$), with each hole occupying around 80% of the inter-ring gap. A practical upper bound follows: hole diameter should remain below approximately half the inter-ring spacing.

4.6 Comparison with Krasteva and Papazov (2002)

Krasteva and Papazov [4] showed that linearly arranged millimetre-scale slots could reduce K by 71%. The NSGA-II optimisation here achieved 9.0% and the ring simulation 2.3% at the 0.4 mm baseline. Both

studies use the same computational domain and confirm the same underlying principle: insulating regions redistribute current away from the electrode periphery [4, 7]. This work extends their approach by adding a systematic multi-objective optimisation framework, a geometry that can be manufactured with standard laser-cutting equipment, and quantitative sensitivity data relating each geometric parameter to performance.

4.7 Routes to Further Improvement

The clearest route to improvement is larger holes. Diameters of 1.0–1.5 mm access the steep region of the sensitivity curve, as confirmed by the -5.8% result at 1.5 mm. The laser cutter can produce these sizes with reduced thermal dimensional error, though ring spacing would need to increase to 8–10 mm to preserve clearance.

Reducing the perforation start radius r_{perf} is a second option. The optimisation’s preference for r_{perf} below 10 mm suggests extending holes towards the centre could yield meaningful gains, but would require better coating control (e.g., screen printing) or alternative hole geometries preserving continuous conduction paths.

Back electrode length operates independently. Increasing the back pad from 100 to 140 mm reduced K by 12% while heart-wall mean $|J|$ rose only modestly. A 140 mm back pad was selected for the prototype, giving roughly twice the baseline area. Combining pad length and perforation improvements could produce cumulative gains, though the interaction would need its own optimisation.

4.8 Computational Limitations

The model uses a two-dimensional cross-section, which captures the radial redistribution of current but cannot represent the angular distribution of holes around each ring; a full three-dimensional simulation would be needed for this. Tissue is modelled as a homogeneous bulk conductor, appropriate for comparative design evaluation but meaning that absolute current density values should not be interpreted as *in vivo* predictions. Mesh resolution differed between the NSGA-II search and the ring simulation, accounting for the difference in solid baseline K (2.56 versus 2.82), and the back pad sweep used a continuous impedance model rather than resolved perforations, so its K values are not directly comparable. These simplifications do not affect the relative ranking of designs, which is the basis for all conclusions drawn. Future work could address the mesh consistency issue and accelerate the optimi-

sation using surrogate models [13].

4.9 Conductive Electrode Layer

The conductive metallic layer was central to the aim of reducing the edge effect by redistributing current away from the pad periphery. This is consistent with previous work showing that insulating features can shift current away from electrode edges and that impedance-gradient electrode designs may reduce skin irritation during transthoracic stimulation [4, 7]. The computational modelling successfully demonstrated the current concentration associated with standard electrode geometries and showed that the perforated electrode concept improved current uniformity *in silico*. but the final design also had to be manufacturable.

This required several compromises. The optimisation favoured larger holes and a smaller perforation start radius, but the final ring design used 0.4 mm holes because this matched the minimum feature size of the CO2 laser cutter. A solid central region was also retained to support more reliable Ag/AgCl coating and hydrogel contact, even though the optimisation suggested that extending perforations closer to the centre would improve uniformity. The conductive layer therefore represented a compromise between electrical performance and practical fabrication rather than a direct implementation of the computational optimum.

The fabrication route introduced further uncertainty. Laser cutting very small perforations may have caused local melting, edge deformation, or partial burning of the Mylar, which could make the final hole geometry differ from the CAD design. Since the intended resistance gradient depends on precise control of perforation size and spacing, even small dimensional changes may alter current pathways and weaken the expected edge-effect mitigation. Manual application of the Ag/AgCl paste would not provide precise control over coating thickness, while layer alignment and adhesion within the full pad assembly could further affect electrical consistency. Compared with commercially manufactured standard pads, this prototype route is therefore more susceptible to fabrication-related variation.

Overall, the conductive electrode layer work showed that the optimised concept could be translated into a laser-cut substrate, but it also highlighted the gap between computationally favourable designs and practically manufacturable ones. Due to limited time for assembly and experimental validation, it remains uncertain how much of the predicted improvement would be preserved in the final pad. Future work should focus on larger and more precise perforations, more consistent conductive coating, and tighter quality control so that the design can be assessed more reliably against

the computational predictions.

4.10 Hydrogel Material Selection

The hydrogel layer is important because it determines the quality of electrical coupling between the electrode and the skin. By conforming to surface irregularities, it can reduce contact impedance and promote more uniform current distribution across the skin-electrode interface. This is particularly important in external cardiac pacing, where uneven current density can increase pain and discomfort. Pad performance therefore depends not only on the conductivity of the metallic layer, but also on the interfacial properties of the hydrogel.

Mindray radiolucent electrodes were used as the control; however, their hydrogel formulation is proprietary and not publicly disclosed. This was a significant limitation in the material selection process, as pacing hydrogels must balance conductivity, adhesion, biocompatibility, and mechanical stability. Without access to the control composition, it was difficult to determine which properties should be prioritised to reduce peripheral current concentration while maintaining skin contact. As a result, most design optimisation focused on the Ag/AgCl layer.

Further literature review led to the selection of PEDOT:PSS as the hydrogel material because of its high conductivity and ease of preparation. This provided a rational basis for prototype development and established a suitable starting point for future testing. However, it lacks strong intrinsic adhesive properties, which would limit conformal contact with skin and may increase interfacial impedance in practice [14]. For this reason, future work will explore blending PEDOT:PSS with PDMAPS, a biocompatible zwitterionic polymer with self-adhesive properties, in order to improve skin contact while maintaining good conductivity.

PDMAPS may improve conformal skin contact by increasing PEDOT⁺ chain mobility and promoting favourable interactions with the hydrated skin surface, which could reduce interfacial impedance without significantly compromising conductivity [14]. However, this was not explored experimentally within the scope of the project. Limited laboratory access, regulatory requirements, and the 24-hour gelation time for PEDOT:PSS reduced the time available for hydrogel development, so alternative formulations could not be synthesised or optimised before the report deadline. Incorporating PDMAPS would also reduce the relative concentration of PEDOT:PSS, which may introduce a trade-off between adhesion and electrical performance. Further testing would therefore be needed to determine whether this trade-off affects current distribution and overall pad performance.

Despite this, the hydrogel selection process provided a clear direction for future development. The project therefore moved beyond electrode geometry alone and identified both a suitable conductive layer design and a basis for future hydrogel optimisation.

4.11 Testing and Validation

The computational results suggested that the perforated design would reduce edge-current concentration compared with the standard pad. To test this experimentally, a bench validation protocol was also developed using a gelatin-saline phantom with temperature-sensitive visualisation. This was an important outcome of the project, as it defined a practical method for comparing hotspot formation in the redesigned pad against the standard Mindray control.

Although this validation was not completed before the report deadline because delays in delivery of the Ag/AgCl paste and hydrogel prevented final prototype assembly, the project still established a clear next step for translation from modelling to experiment. The proposed gelatin-saline phantom with temperature-sensitive ink remains a useful validation method because it would allow spatial visualisation of heat distribution across the pad-phantom interface. This is directly relevant to the edge-effect problem, as it would show whether heating begins preferentially at the pad periphery.

However, the method also has important limitations. The temperature-sensitive ink provides only a surface-level indication of heating and does not capture sub-surface thermal gradients, where tissue damage is more clinically relevant. In addition, its 30°C transition acts as a threshold response rather than a continuous temperature measurement, limiting its ability to distinguish between different degrees of thermal rise.

Future validation should therefore focus not only on comparing the perforated pad with the Mindray control, but also on improving the quality of the measurement itself. Supplementing or replacing the surface ink with fibre-optic probes would allow depth-resolved and quantitative temperature monitoring. Once the full prototype is assembled, this would provide a more reliable assessment of whether the reduction in edge effects predicted by the computational model is preserved in the manufactured device.

In summary, this project has established through computational modelling that a perforated impedance-gradient electrode design produces a measurable reduction in edge-current density compared with a standard solid electrode. The multi-objective optimisation framework successfully identified perforation start radius, hole radius, and hole count as the key geomet-

ric parameters, and the resulting Pareto front provides a quantitative basis for selecting designs that balance uniformity against peak current density. These findings directly informed the manufacturable concentric-ring electrode design (Figure 3), which achieved a 2.3% reduction in K at the 0.4mm baseline hole diameter, with simulations indicating that larger perforations (1.0–1.5 mm) could yield improvements of up to 5.8%. The immediate next steps are phantom testing using the established protocol, followed by animal testing to assess in vivo performance.

5 References

- [1] Sokup B, Sheils E, and Ahmed I. Transcutaneous pacing. *StatPearls*, 2026. Internet; cited 2026 Apr 12. Available from: <https://www.ncbi.nlm.nih.gov/sites/books/NBK519567/>.
- [2] M. J. Singleton, M. H. Schoenfeld, P. D. Bhavé, E. H. Beaty, and S. P. Whalen. Third-degree burns caused by transcutaneous pacing for third-degree heart block. *HeartRhythm Case Reports*, 2020. Internet; cited 2026 Apr 12. Available from: [https://www.heartrhythmcasereports.com/article/S2214-0271\(20\)30092-0/fulltext](https://www.heartrhythmcasereports.com/article/S2214-0271(20)30092-0/fulltext).
- [3] Non-invasive transcutaneous pacing. *Philips Application Note*. Internet; cited 2026 Apr 12. Available from: <https://www.documents.philips.com/assets/20170523/d18d46f2d0fe4a048995a77c0154d4e8.pdf>.
- [4] V. T. Krasteva and S. P. Papazov. Estimation of current density distribution under electrodes for external defibrillation. *BioMedical Engineering OnLine*, 2002. Internet; cited 2026 Apr 12. Available from: <https://link.springer.com/article/10.1186/1475-925X-1-7>.
- [5] B. Wang, A. Petrossians, and J. D. Weiland. Reduction of edge effect on disk electrodes by optimized current waveform. *IEEE Transactions on Biomedical Engineering*, 2014. Internet; cited 2026 Apr 12. Available from: <https://pmc.ncbi.nlm.nih.gov/articles/PMC4255912/>.
- [6] X. Muschart. Burns to be alive: a complication of transcutaneous cardiac stimulation. *Critical Care*, 2014. Internet; cited 2026 Apr 12. Available from: <https://pmc.ncbi.nlm.nih.gov/articles/PMC4236425/>.
- [7] P. F. Meyer, P. D. Gadsby, D. V. Sickel, W. E. Schoenlein, K. S. Foster, and G. P. Graber. Impedance-gradient electrode reduces skin irritation induced by transthoracic defibrillation. *Medical & Biological Engineering & Computing*, 2005. Internet; cited 2026 Apr 10.
- [8] K. Deb, A. Pratap, S. Agarwal, and T. Meyarivan. A fast and elitist multiobjective genetic algorithm: Nsga-ii. *IEEE Transactions on Evolutionary Computation*, 6(2):182–197, 2002. [Internet; cited 2026 Apr 10].
- [9] T. P. DeMonte, P. D. Gadsby, P. F. Meyer, and M. L. Joy. Measurement of edge effects in automatic external defibrillation electrodes using current density imaging. In *Proceedings of the 13th Scientific Meeting & Exhibition of the International Society for Magnetic Resonance in Medicine*, page 957, Miami Beach, FL, May 7–13 2005.
- [10] M. A. E. Andersen, O. S. Neves, and E. S. L. Lima. Global sensitivity analysis of bioelectromagnetic models: application to transcranial direct current stimulation. *Journal of Neural Engineering*, 17(4):046028, 2020.
- [11] R. T. Marler and J. S. Arora. Survey of multi-objective optimization methods for engineering. *Structural and Multidisciplinary Optimization*, 26(6):369–395, 2004.
- [12] A. M. D. Jorge, A. C. de Oliveira, and A. C. de Souza. Finite element analysis of current density and electric field in transcutaneous electrical nerve stimulation. *Medical Engineering & Physics*, 38(3):282–289, 2016.
- [13] A. Forrester, A. Sobester, and A. Keane. *Engineering Design via Surrogate Modelling: A Practical Guide*. Wiley, 2008.
- [14] Xiaoxue Lin, Rui Chen, Zeping Ou, Wen Sun, Lei Wang, Zhihan Zeng, Hanlin Cheng, Kuan Sun, and Jianyong Ouyang. Highly stretchable and self-adhesive pedot:pss blends via coulombic repulsion-induced weakening of interchain interaction. *Chemical Engineering Journal*, 2026.
- [15] Y. Kim, J. Ben Fahy, and Bradley J. Tupper. Optimal electrode designs for electrosurgery, defibrillation, and external cardiac pacing. *IEEE Transactions on Biomedical Engineering*, BME-33(9):845–853, 1986.
- [16] D. Panescu, J. G. Webster, W. J. Tompkins, and R. A. Stratbucker. Optimisation of transcutaneous cardiac pacing by three-dimensional finite element modelling of the human thorax. 33:769–775, 1995.
- [17] P.F. Meyer and P.D. Gadsby. Electrode

chemistry of prolonged external pacing. In *IEEE 30th Annual Northeast Bioengineering Conference, 2004. Proceedings of the*, pages 33–34, 2004.

- [18] L. A. Garcia, L. A. Pagan-Carlo, M. S. Stone, and R. E. Kerber. High perimeter impedance defibrillation electrodes reduce skin burns in transthoracic cardioversion. *American Journal of Cardiology*, 82(9):1125–1127, 1998.

6 Appendix

The main departure from our original project plan, shown in Figure 10, was that full prototype assembly and bench validation were delayed beyond the report deadline. However, the project achieved its main technical development goals within the deadline. We finalised the perforated pacing pad design, completed the computational modelling, produced the laser-cut Mylar electrode layer, and established the manufacturing workflow for final assembly. These outcomes provided a strong basis for evaluating the design concept and demonstrating substantial technical progress. The delay was mainly due to procurement issues. Supplier lead times for the Ag/AgCl ink and hydrogel, together with purchasing approval and delivery arrangements, meant that these materials arrived later than expected. As a result, we revised the project timeline and focused the report on the completed design, modelling, and manufacturing preparation, with bench testing followed by animal testing forming the next stage of the project.

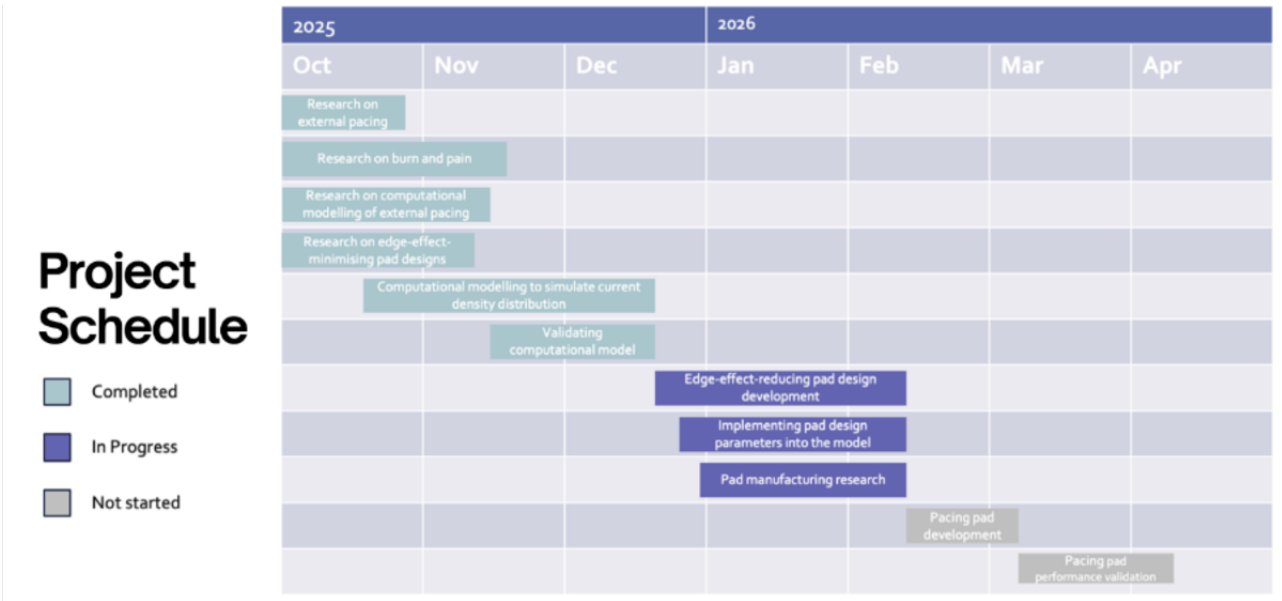


Figure 10: Original project Gantt chart showing the planned timeline for research, modelling, design development, manufacturing, and validation.

A key lesson from this project was the importance of building contingency into procurement planning for specialist materials. Although the technical work progressed as expected, delays in obtaining Ag/AgCl ink and hydrogel affected the timing of final prototype assembly. In future projects, materials with uncertain supplier lead times should be identified and ordered much earlier. Project schedules should also include buffer time for approval, shipping, and delivery issues. This would reduce the risk of external delays affecting the final stages of manufacture and testing.

Although prototype assembly was delayed, the project was still able to move forward because progress had already been made in other areas. The design had been finalised, the computational modelling completed, and the Mylar electrode layer laser cut before the missing materials became the main constraint. This allowed the group to continue producing meaningful outcomes despite not completing full experimental validation. The project showed the value of organising work so that modelling, design, and manufacturing preparation can continue in parallel, even when one part of the project is delayed.

The project highlighted the importance of responding realistically when the original plan cannot be completed as intended. Once it became clear that the materials would not arrive before the deadline, we informed the supervisor, revised the timeline, and refocused the report on the computational findings and manufacturing workflow. This allowed us to base the report on completed work while acknowledging that experimental validation had not yet been carried out. An important lesson was that effective project management involves not only following a plan, but also adjusting scope and priorities when needed.

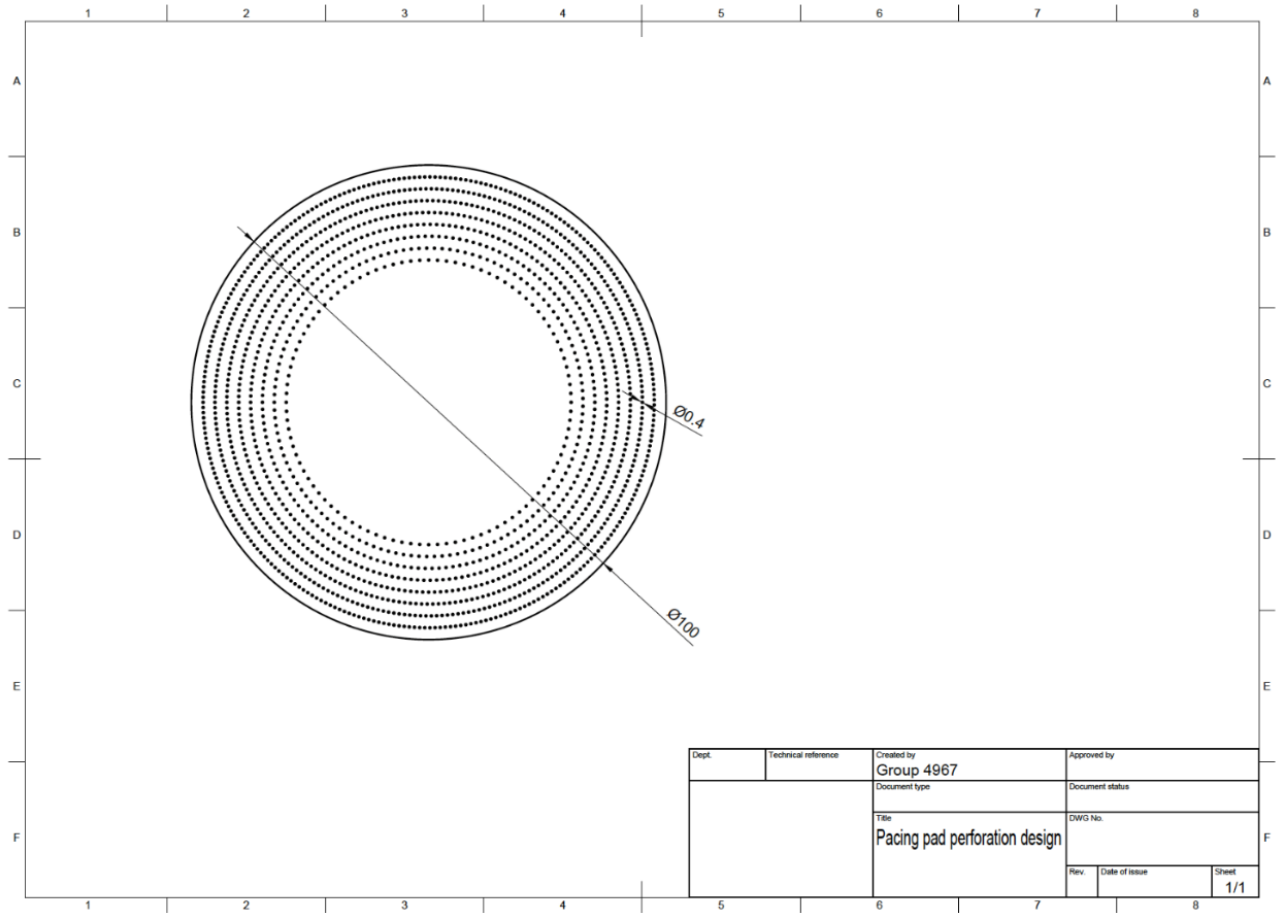


Figure 11: Technical drawing of the perforated conductive layer design for the front pacing pad, created in Autodesk Fusion 360.

6.1 Code

6.1.1 NSGA-II Optimisation

```
function perforatedElectrodeNSGA2()
% NSGA-II optimisation of perforated Ag/AgCl defibrillation electrode.
% Decision variables: hole radius [m], holes per side, r_perf [m]
% Objectives: minimise K (non-uniformity) and peak |J| in tissue
% Ref: Krasteva & Papazov (2002), BioMedical Engineering OnLine.

close all; clc;

fprintf('=====\n');
fprintf(' NSGA-II OPTIMISATION: PERFORATED Ag/AgCl ELECTRODE\n');
fprintf('=====\n\n');

%% 1. Parameters and geometry
p = defaultParameters();
[p.dl, p.externalEdges, p.elec1Edge, p.elec2Edge] = buildGeometry(p);

%% 2. Variable bounds
lb = [0.0002, 2, 0.005];
ub = [0.0025, 30, 0.045];
nVars = 3;

fprintf('Bounds:\n');
```

```

fprintf(' hole radius: [%.1f, %.1f] mm\n', lb(1)*1e3, ub(1)*1e3);
fprintf(' holes per side: [%d, %d]\n', lb(2), ub(2));
fprintf(' r_perf: [%.1f, %.1f] mm\n\n', lb(3)*1e3, ub(3)*1e3);

%% 3. NSGA-II setup
popSize = 60;
maxGen = 40;
opts = optimoptions('gamultiobj', ...
    'PopulationSize', popSize, 'MaxGenerations', maxGen, ...
    'CrossoverFraction', 0.8, 'ParetoFraction', 0.35, ...
    'FunctionTolerance', 1e-4, 'Display', 'iter', 'UseParallel', false);
intcon = 2;

%% 4. Run optimisation
allX = []; allF = [];

function f = evaluateAndStore(x)
    f = evaluateDesign(x, p);
    allX = [allX; x];
    allF = [allF; f];
end

fprintf('Starting NSGA-II (pop=%d, gen=%d) ...\n', popSize, maxGen);
fprintf('-----\n');
tStart = tic;
[xPareto, fPareto] = gamultiobj(@evaluateAndStore, nVars, ...
    [], [], [], [], lb, ub, [], intcon, opts);
tTotal = toc(tStart);
fprintf('-----\n');
fprintf('Done in %.1f min | %d Pareto, %d total evals\n\n', ...
    tTotal/60, size(xPareto,1), size(allX,1));

%% 5. Post-process
[~, si] = sort(fPareto(:,1));
xPareto = xPareto(si,:);
fPareto = fPareto(si,:);

validMask = allF(:,1) < 50 & allF(:,2) < 50;
allXv = allX(validMask,:);
allFv = allF(validMask,:);

fprintf('PARETO FRONT:\n');
fprintf(' %-4s %-8s %-6s %-10s | %-8s %-10s\n', ...
    '#', 'r(mm)', 'n/s', 'r_perf(mm)', 'K', 'peakJ');
for ii = 1:size(xPareto,1)
    fprintf(' %-4d %-8.2f %-6d %-10.1f | %-8.2f %-10.4f\n', ...
        ii, xPareto(ii,1)*1e3, round(xPareto(ii,2)), ...
        xPareto(ii,3)*1e3, fPareto(ii,1), fPareto(ii,2));
end

[bestKIdx, kneeIdx, bestJIdx] = identifyKeySolutions(fPareto);
labels = {'Best K', 'Knee', 'Best peakJ'};
for kk = 1:3
    ii = [bestKIdx, kneeIdx, bestJIdx]; ii = ii(kk);
    fprintf(' %-10s: K=%.2f peakJ=%.2f (r=%.2fmm n=%d rp=%.1fmm)\n', ...
        labels{kk}, fPareto(ii,1), fPareto(ii,2), ...
        xPareto(ii,1)*1e3, round(xPareto(ii,2)), xPareto(ii,3)*1e3);
end

%% 6. Solid baseline
fprintf('\nSolid baseline: ');
[~, ~, Ks, ~, pJs] = solveElectrode(0, 0, 0.025, p, false);
fprintf('K=%.2f, peakJ=%.2f\n\n', Ks, pJs);

```

```

%% 7. High-res re-solve
fprintf('Re-solving at fine mesh ...\n');
xBestJ = xPareto(bestJIdx,:);
[resSolid, rzSolid, KsHR, ~, pJsHR] = solveElectrode(0, 0, 0.025, p, true);
[resBestJ, rzBestJ, KbjHR, ~, pJbjHR] = ...
    solveElectrode(xBestJ(1), round(xBestJ(2)), xBestJ(3), p, true);
fprintf(' Solid: K=%.2f, peakJ=%.2f\n', KsHR, pJsHR);
fprintf(' BestJ: K=%.2f, peakJ=%.2f\n\n', KbjHR, pJbjHR);

%% 8. Line profiles (0.5 mm below interface)
evalZ = p.d + 0.0005;
[distSolid, JlineSolid] = extractLineProfile(rzSolid, p, evalZ);
[distPerf, JlinePerf] = extractLineProfile(rzBestJ, p, evalZ);

%% 9. Figures
cfg = plotDefaults();
plotParetoFront(allXv, allFv, xPareto, fPareto, bestKIdx, kneeIdx, bestJIdx, Ks, pJs, cfg);
plotCurrentMaps(resSolid, resBestJ, p, KsHR, KbjHR, cfg);
plotZoomedMaps(rzSolid, rzBestJ, p, cfg);
plotLineProfiles(distSolid, JlineSolid, distPerf, JlinePerf, p, xBestJ, cfg);
plotSensitivity(allXv, allFv, xPareto, fPareto, Ks, pJs, cfg);

%% 10. Save
save('nsga2_perforated_results.mat', ...
    'xPareto', 'fPareto', 'allXv', 'allFv', 'p', ...
    'Ks', 'pJs', 'KsHR', 'pJsHR', 'KbjHR', 'pJbjHR', ...
    'xBestJ', 'bestKIdx', 'bestJIdx', 'kneeIdx');
fprintf('Results saved. All figures generated.\n');
end

%% ---- defaultParameters ----
function p = defaultParameters()
    p.domainWidth = 0.50; % m
    p.domainHeight = 0.25; % m
    p.heartRadius = 0.045; % m
    p.heartX = 0.25; p.heartZ = 0.095;
    p.probeX = 0.25; p.probeZ = 0.05;
    p.sigma_tissue = 0.2; % S/m
    p.sigma_heart = 0.5;
    p.Vapp = 10; % V
    p.padWidth = 0.10; % m
    p.pad1X = 0.25; p.pad2X = 0.25;
    p.d = 0.001; % electrode thickness
    p.rho_metal = 5; % surface resistivity
    p.z_metal = p.rho_metal * p.d;
    p.sigma_metal = p.d / p.z_metal;
    p.sigma_hole = 1e-8; % insulating
    p.pad1L = p.pad1X - p.padWidth/2;
    p.pad1R = p.pad1X + p.padWidth/2;
    p.pad2L = p.pad2X - p.padWidth/2;
    p.pad2R = p.pad2X + p.padWidth/2;
    p.R_pad = p.padWidth / 2;
end

%% ---- buildGeometry ----
function [dl, externalEdges, elec1Edge, elec2Edge] = buildGeometry(p)
% Rectangle (torso) + circle (heart). Bottom and top edges split at
% electrode pad boundaries so BCs can be applied per-edge.
vertices = [0,0; p.pad1L,0; p.pad1R,0; p.domainWidth,0;
            p.domainWidth,p.domainHeight; p.pad2R,p.domainHeight;

```

```

        p.pad2L,p.domainHeight; 0,p.domainHeight];
nv = size(vertices,1);
pgon = [2; nv; vertices(:,1); vertices(:,2)];
circ = [1; p.heartX; p.heartZ; p.heartRadius; zeros(6,1)];
maxLen = max(length(pgon),length(circ));
pgon(end+1:maxLen) = 0; circ(end+1:maxLen) = 0;

[dl,~] = decsg([pgon,circ], 'P1+C1', char('P1','C1'));

numEdges = size(dl,2);
externalEdges = [];
for eID = 1:numEdges
    if dl(6,eID)==0 || dl(7,eID)==0
        externalEdges(end+1) = eID; %#ok<AGROW>
    end
end

elec1Edge = 2; elec2Edge = 6;
tol = 1e-6;
for eID = externalEdges
    mx = (dl(2,eID)+dl(3,eID))/2;
    my = (dl(4,eID)+dl(5,eID))/2;
    if abs(my)<tol && mx>p.pad1L-tol && mx<p.pad1R+tol
        elec1Edge = eID;
    end
    if abs(my-p.domainHeight)<tol && mx>p.pad2L-tol && mx<p.pad2R+tol
        elec2Edge = eID;
    end
end
fprintf('Geometry: %d edges, elec1=%d, elec2=%d\n\n', numEdges, elec1Edge, elec2Edge);
end

%% ---- solveElectrode ----
function [res, resZoom, K, JAtProbe, peakJtissue] = ...
    solveElectrode(holeR, nH, rPerf, p, highRes)
% Solve -div(sigma*grad(V))=0 for a given perforation pattern.
% holeR=0 or nH=0 gives solid electrode.

% Compute hole positions
if holeR <= 0 || nH < 1
    useHoles = false; holeLArr = []; holeRArr = [];
    rPerfUsed = rPerf;
else
    useHoles = true;
    rPerfUsed = max(0.002, min(rPerf, p.R_pad - 0.002));
    L = p.R_pad - rPerfUsed;
    if L < 0.002
        useHoles = false; holeLArr = []; holeRArr = [];
    else
        spacing = L / (nH + 1);
        actualR = min(holeR, spacing * 0.40); % 40% cap prevents overlap
        centresR = p.pad1X + rPerfUsed + spacing*(1:nH)';
        centresL = p.pad1X - rPerfUsed - spacing*(1:nH)';
        allCentres = [centresL; centresR];
        holeLArr = allCentres - actualR;
        holeRArr = allCentres + actualR;
    end
end

function tf = inHole(xi)
    if ~useHoles || abs(xi - p.pad1X) < rPerfUsed
        tf = false;
    end
end

```



```

        else
            tf = any((xi >= holeLArr) & (xi <= holeRArr));
        end
    end

% PDE model
mdl = createpde();
geometryFromEdges(mdl, p.dl);
if highRes
    generateMesh(mdl, 'Hmax',0.0012, 'Hmin',0.0003, 'GeometricOrder','linear');
else
    generateMesh(mdl, 'Hmax',0.002, 'Hmin',0.0005, 'GeometricOrder','linear');
end

% Conductivity: tissue/heart/metal/hole depending on location
function sig = condCoeff(region, ~)
    x = region.x; z = region.y;
    sig = p.sigma_tissue * ones(size(x));
    for ii = 1:numel(x)
        xi = x(ii); zi = z(ii);
        if (xi-p.heartX)^2 + (zi-p.heartZ)^2 <= p.heartRadius^2
            sig(ii) = p.sigma_heart;
        elseif zi <= p.d && xi >= p.pad1L && xi <= p.pad1R
            if inHole(xi), sig(ii) = p.sigma_hole;
            else, sig(ii) = p.sigma_metal; end
        elseif zi >= p.domainHeight-p.d && xi >= p.pad2L && xi <= p.pad2R
            if inHole(xi), sig(ii) = p.sigma_hole;
            else, sig(ii) = p.sigma_metal; end
        end
    end
end

% BCs: V=0 ground, V=Vapp active, dV/dn=0 elsewhere
specifyCoefficients(mdl, 'm',0, 'd',0, 'c',@condCoeff, 'a',0, 'f',0);
applyBoundaryCondition(mdl, 'dirichlet', 'Edge', p.elec1Edge, 'u', 0);
applyBoundaryCondition(mdl, 'dirichlet', 'Edge', p.elec2Edge, 'u', p.Vapp);
for eID = setdiff(p.externalEdges, [p.elec1Edge, p.elec2Edge])
    applyBoundaryCondition(mdl, 'neumann', 'Edge', eID, 'g', 0, 'q', 0);
end

% Solve
result = solvepde(mdl);
nodeX = mdl.Mesh.Nodes(1,:);
nodeZ = mdl.Mesh.Nodes(2,:);
[gradX, gradZ] = evaluateGradient(result, nodeX, nodeZ);

% Nodal conductivities for J = -sigma*grad(V)
sigN = p.sigma_tissue * ones(size(nodeX));
for ii = 1:numel(nodeX)
    xi = nodeX(ii); zi = nodeZ(ii);
    if (xi-p.heartX)^2+(zi-p.heartZ)^2 <= p.heartRadius^2
        sigN(ii) = p.sigma_heart;
    elseif zi <= p.d && xi >= p.pad1L && xi <= p.pad1R
        if inHole(xi), sigN(ii) = p.sigma_hole;
        else, sigN(ii) = p.sigma_metal; end
    elseif zi >= p.domainHeight-p.d && xi >= p.pad2L && xi <= p.pad2R
        if inHole(xi), sigN(ii) = p.sigma_hole;
        else, sigN(ii) = p.sigma_metal; end
    end
end

Jmag = sqrt((-sigN.*gradX).^2 + (-sigN.*gradZ).^2);

% Interpolate onto regular grids

```

```

nGx = 200; nGz = 100;
if highRes, nGx = 250; nGz = 125; end
[gX,gZ] = meshgrid(linspace(0,p.domainWidth,nGx), linspace(0,p.domainHeight,nGz));
JG = griddata(nodeX,nodeZ,Jmag,gX,gZ);
VG = griddata(nodeX,nodeZ,result.NodalSolution,gX,gZ);
res = struct('J',JG, 'V',VG, 'X',gX, 'Z',gZ);

if highRes
    [zX,zZ] = meshgrid(linspace(p.pad1L-0.005,p.pad1R+0.005,500), linspace(0,0.025,200));
    resZoom = struct('J',griddata(nodeX,nodeZ,Jmag,zX,zZ), ...
        'V',griddata(nodeX,nodeZ,result.NodalSolution,zX,zZ), 'X',zX, 'Z',zZ);
else
    resZoom = struct('J',[], 'V',[], 'X',[], 'Z',[]);
end

% K = Jmax/Jmin along line 0.5mm below interface
evalZ = p.d + 0.0005;
[~,zi] = min(abs(gZ(:,1) - evalZ));
xm = (gX(1,:) >= p.pad1L) & (gX(1,:) <= p.pad1R);
Js = JG(zi,xm); Js = Js(~isnan(Js) & Js>0);
if ~isempty(Js), K = max(Js)/min(Js); else, K = NaN; end

% Probe |J|
JAtProbe = interp2(gX,gZ,JG,p.probeX,p.probeZ,'linear');
if isnan(JAtProbe)
    dd = (gX-p.probeX).^2 + (gZ-p.probeZ).^2;
    dd(isnan(JG)) = inf;
    [~,idx] = min(dd(:)); JAtProbe = JG(idx);
end

% Peak |J| in tissue (excl. heart and electrode layers)
heartM = ((gX-p.heartX).^2+(gZ-p.heartZ).^2) <= p.heartRadius^2;
intfM = (gZ <= p.d) | (gZ >= p.domainHeight-p.d);
Jt = JG(~heartM & ~intfM & ~isnan(JG));
if ~isempty(Jt), peakJtissue = max(Jt); else, peakJtissue = NaN; end
end

%% ---- evaluateDesign ----
function f = evaluateDesign(x, p)
% Objective wrapper for gamultiobj. Returns [100,100] on failure.
try
    [~,~,K,~,peakJ] = solveElectrode(x(1), round(x(2)), x(3), p, false);
    if isnan(K) || isnan(peakJ), f = [100,100];
    else, f = [K, peakJ]; end
catch
    f = [100, 100];
end
end

%% ---- identifyKeySolutions ----
function [bestKIdx, kneeIdx, bestJIdx] = identifyKeySolutions(fPareto)
[~, bestKIdx] = min(fPareto(:,1));
[~, bestJIdx] = min(fPareto(:,2));
% Knee: closest to origin in normalised space
fNorm = (fPareto - min(fPareto)) ./ (max(fPareto) - min(fPareto) + eps);
[~, kneeIdx] = min(sqrt(fNorm(:,1).^2 + fNorm(:,2).^2));
end

%% ---- extractLineProfile ----
function [dist, Jline] = extractLineProfile(resZoom, p, evalZ)

```

```

    [~,zIdx] = min(abs(resZoom.Z(:,1) - evalZ));
    xLine = resZoom.X(1,:);
    Jrow = resZoom.J(zIdx,:);
    distFull = xLine - p.pad1X;
    mask = abs(distFull) <= p.R_pad + 0.005;
    dist = distFull(mask); Jline = Jrow(mask);
end

%% ---- plotDefaults ----
function cfg = plotDefaults()
    cfg.fs = 10; cfg.fn = 'Helvetica';
    cfg.colAll = [0.75 0.75 0.75];
    cfg.setAx = @(ax) set(ax, 'FontSize',10, 'FontName','Helvetica', ...
        'TickDir','out', 'LineWidth',0.5, 'Box','on');
end

%% ---- plotParetoFront ----
function plotParetoFront(allXv, allFv, xP, fP, bK, kn, bJ, Ks, pJs, cfg)
    figure('Units','cm','Position',[2 4 17.4 8],'Color','w','PaperPositionMode','auto');
    sgtitle('NSGA-II Results','FontSize',cfg.fs+1,'FontName',cfg.fn,'FontWeight','bold');

    subplot(1,2,1);
    plot(allFv(:,1), allFv(:,2), '.', 'Color',cfg.colAll, 'MarkerSize',4); hold on;
    plot(fP(:,1), fP(:,2), 'ko', 'MarkerSize',5, 'MarkerFaceColor','k');
    plot(fP(bK,1),fP(bK,2),'s','Color',[0 .5 0],'MarkerSize',10,'MarkerFaceColor',[0 .7 0],'LineWidth',
        1.2);
    plot(fP(bJ,1),fP(bJ,2),'d','Color',[.6 0 0],'MarkerSize',10,'MarkerFaceColor',[.9 .2 .2],'
        LineWidth',1.2);
    plot(fP(kn,1),fP(kn,2),'^','Color',[0 0 .6],'MarkerSize',10,'MarkerFaceColor',[.2 .2 .9],'
        LineWidth',1.2);
    plot(Ks, pJs, 'kx', 'MarkerSize',12, 'LineWidth',2.5); hold off;
    xlabel('K'); ylabel('Peak |J| (A/m^2)');
    title('(a) Objective space'); legend('All','Pareto','Best K','Best peakJ','Knee','Solid','Location',
        'best','FontSize',7,'Box','off');
    cfg.setAx(gca);

    subplot(1,2,2);
    scatter3(allXv(:,1)*1e3, allXv(:,2), allXv(:,3)*1e3, 10, allFv(:,1), 'filled','MarkerFaceAlpha',
        0.3);
    colormap(gca,parula(64)); cb=colorbar; ylabel(cb,'K');
    hold on;
    scatter3(xP(:,1)*1e3, xP(:,2), xP(:,3)*1e3, 40, fP(:,1), 'filled','MarkerEdgeColor','k','LineWidth',
        0.8);
    hold off;
    xlabel('Hole radius (mm)'); ylabel('Holes/side'); zlabel('r_{perf} (mm)');
    title('(b) Decision space'); view(135,25); cfg.setAx(gca);
end

%% ---- plotSensitivity ----
function plotSensitivity(allXv, allFv, xP, fP, Ks, pJs, cfg)
    varNames = {'Hole radius (mm)','Holes/side','r_{perf} (mm)'};
    varScales = [1000, 1, 1000];
    objNames = {'K','Peak |J| (A/m^2)'};
    baselines = [Ks, pJs];

    figure('Units','cm','Position',[1 1 17.4 17],'Color','w','PaperPositionMode','auto');
    sgtitle('Parameter Sensitivity','FontSize',cfg.fs+1,'FontName',cfg.fn,'FontWeight','bold');

    for row = 1:2
        for col = 1:3

```

```

        subplot(2,3,(row-1)*3+col);
        scatter(allXv(:,col)*varScales(col), allFv(:,row), 8, cfg.colAll, 'filled'); hold on;
        scatter(xP(:,col)*varScales(col), fP(:,row), 20, 'k', 'filled');
        yline(baselines(row),'r--','LineWidth',0.8); hold off;
        xlabel(varNames{col}); ylabel(objNames{row});
        title(sprintf('(%s)', char('a'-1+(row-1)*3+col)));
        cfg.setAx(gca);
    end
end
end

%% ---- plotCurrentMaps ----
function plotCurrentMaps(resSolid, resBestJ, p, KsHR, KbJHR, cfg)
    Jall = [resSolid.J(:); resBestJ.J(:)];
    clim = [0, prctile(Jall(~isnan(Jall)), 99)];
    th = linspace(0,2*pi,150);
    hx = p.heartX*100 + p.heartRadius*100*cos(th);
    hz = p.heartZ*100 + p.heartRadius*100*sin(th);

    figure('Units','cm','Position',[2 3 17.4 8],'Color','w','PaperPositionMode','auto');
    sgtitle('|J|: Solid vs Optimised','FontSize',cfg.fs+1,'FontName',cfg.fn,'FontWeight','bold');

    for kk = 1:2
        ax = subplot(1,2,kk);
        if kk==1, dat=resSolid; Kv=KsHR; lbl='(a) Solid';
        else, dat=resBestJ; Kv=KbJHR; lbl='(b) Best peakJ'; end
        imagesc([0 50],[0 25],dat.J); colormap(ax,jet(256)); caxis(clim);
        hold on; plot(hx,hz,'w-','LineWidth',1);
        plot(p.probeX*100,p.probeZ*100,'w+', 'MarkerSize',6,'LineWidth',1.2); hold off;
        xlabel('x (cm)'); ylabel('z (cm)');
        title(sprintf('%s, K=%.2f',lbl,Kv));
        axis equal tight; xlim([0 50]); ylim([0 25]);
        set(ax,'YDir','normal'); cfg.setAx(ax);
    end
    cb = colorbar('Position',[0.925 0.17 0.018 0.70]);
    ylabel(cb,'|J| (A/m^2)');
end

%% ---- plotZoomedMaps ----
function plotZoomedMaps(rzSolid, rzBestJ, p, cfg)
    zAll = [rzSolid.J(:); rzBestJ.J(:)];
    zClim = [0, prctile(zAll(~isnan(zAll)), 98)];

    figure('Units','cm','Position',[2 2 17.4 8],'Color','w','PaperPositionMode','auto');
    sgtitle('Zoomed |J| Under Electrode','FontSize',cfg.fs+1,'FontName',cfg.fn,'FontWeight','bold');

    for kk = 1:2
        ax = subplot(1,2,kk);
        if kk==1, rz=rzSolid; lbl='(a) Solid';
        else, rz=rzBestJ; lbl='(b) Best peakJ'; end
        imagesc(rz.X(1,:)*100, rz.Z(:,1)*100, rz.J);
        colormap(ax,jet(256)); caxis(zClim);
        hold on; plot([p.pad1L p.pad1R]*100,[0 0],'w-','LineWidth',2); hold off;
        xlabel('x (cm)'); ylabel('z (cm)'); title(lbl);
        xlim([p.pad1L-0.005,p.pad1R+0.005]*100); ylim([0 2.5]);
        set(ax,'YDir','normal'); cfg.setAx(ax);
    end
    cb = colorbar('Position',[0.925 0.17 0.018 0.70]);
    ylabel(cb,'|J| (A/m^2)');
end

```

```

%% ---- plotLineProfiles ----
function plotLineProfiles(distS, JlineS, distP, JlineP, p, xBJ, cfg)
    % Side-by-side
    figure('Units','cm','Position',[2 1 17.4 8],'Color','w','PaperPositionMode','auto');
    sgtitle('J Distribution Under Electrode','FontSize',cfg.fs+1,'FontName',cfg.fn,'FontWeight','bold'
    );
    subplot(1,2,1);
    plot(distS, JlineS, '-','Color',[0 0 .7], 'LineWidth',1.2);
    xlabel('Distance from centre (m)'); ylabel('|J| (A/m^2)');
    title('(a) Solid'); xlim([-0.06 0.06]); cfg.setAx(gca); grid on;
    subplot(1,2,2);
    plot(distP, JlineP, '-','Color',[.7 0 0], 'LineWidth',1.2);
    xlabel('Distance from centre (m)'); ylabel('|J| (A/m^2)');
    title(sprintf('(b) r=%.1fmm n=%d rp=%.0fmm', xBJ(1)*1e3, round(xBJ(2)), xBJ(3)*1e3));
    xlim([-0.06 0.06]); cfg.setAx(gca); grid on;

    % Overlay
    figure('Units','cm','Position',[2 0 12 8],'Color','w','PaperPositionMode','auto');
    plot(distS*100, JlineS, '-','Color',[0 0 .7], 'LineWidth',1.5); hold on;
    plot(distP*100, JlineP, '-','Color',[.7 0 0], 'LineWidth',1.5);
    xline(-p.R_pad*100,':','Color',[.4 .4 .4], 'LineWidth',0.7);
    xline( p.R_pad*100,':','Color',[.4 .4 .4], 'LineWidth',0.7); hold off;
    xlabel('Distance from centre (cm)'); ylabel('J_S (A/m^2)');
    title('Solid vs Perforated');
    legend('Solid','Perforated','Location','south','FontSize',8,'Box','off');
    xlim([-6 6]); cfg.setAx(gca); grid on;
end

```

6.1.2 Ring Electrode Simulation

```

function ringElectrodeSimulation()
% Simulate the concentric-ring perforated electrode and sweep hole diameters.
% 8 rings x 2 sides = 16 discrete holes in the cross-section.
% Ref: Krasteva & Papazov (2002), BioMedical Engineering OnLine.

close all; clc;

fprintf('=====\n');
fprintf(' RING ELECTRODE SIMULATION: HOLE DIAMETER SWEEP\n');
fprintf('=====\\n\\n');

%% 1. Parameters
p.domainWidth = 0.50;
p.domainHeight = 0.25;
p.heartRadius = 0.045;
p.heartX = 0.25; p.heartZ = 0.095;
p.probeX = 0.25; p.probeZ = 0.05;
p.sigma_tissue = 0.2;
p.sigma_heart = 0.5;
p.Vapp = 10;
p.padWidth = 0.10;
p.pad1X = 0.25; p.pad2X = 0.25;
p.d = 0.001;
p.rho_metal = 5;
p.z_metal = p.rho_metal * p.d;
p.sigma_metal = p.d / p.z_metal;
p.sigma_hole = 1e-8;
p.pad1L = p.pad1X - p.padWidth/2;
p.pad1R = p.pad1X + p.padWidth/2;
p.pad2L = p.pad2X - p.padWidth/2;

```

```

p.pad2R = p.pad2X + p.padWidth/2;
p.R_pad = p.padWidth / 2;

[p.dl, p.externalEdges, p.elec1Edge, p.elec2Edge] = buildRingGeometry(p);

%% 2. Ring layout
ringDiameters_cm = [6.0, 6.5, 7.0, 7.5, 8.0, 8.5, 9.0, 9.5];
ringRadii_m = (ringDiameters_cm / 2) * 1e-2;
nHolesPerRing = [100, 125, 150, 175, 200, 225, 250, 275];
nRings = length(ringRadii_m);

fprintf('Ring layout (8 rings, 5 mm spacing):\n');
for ii = 1:nRings
    fprintf(' Ring %d: d=%.1f cm, r=%.1f mm, %d holes\n', ...
        ii, ringDiameters_cm(ii), ringRadii_m(ii)*1e3, nHolesPerRing(ii));
end

% Hole centres: pad centre +/- each ring radius
holeCentres_m = zeros(2*nRings, 1);
for rr = 1:nRings
    holeCentres_m(2*rr-1) = p.pad1X - ringRadii_m(rr);
    holeCentres_m(2*rr) = p.pad1X + ringRadii_m(rr);
end
fprintf(' 16 holes in cross-section\n\n');

%% 3. Diameter sweep
holeDiameters_mm = [0.1, 0.2, 0.3, 0.4, 0.6, 0.8, 1.0, 1.5, 2.0];
nSweep = length(holeDiameters_mm);

fprintf('Sweep: [%s] mm\n\n', strjoin(arrayfun(@(x) sprintf('%.1f',x), ...
    holeDiameters_mm, 'Uni', false), ', '));

%% 4. Solid baseline
fprintf('Solid baseline: ');
[resSolid, rzSolid, Ks, ~, pJs] = solveWithHoles([], [], p, true);
fprintf('K=%.2f, peakJ=%.4f\n\n', Ks, pJs);

%% 5. Sweep
results = struct('diam_mm',{},'K',{},'peakJ',{},'res',{},'rz',{},'dist',{},'Jline',{});

for ss = 1:nSweep
    holeR_m = holeDiameters_mm(ss) * 1e-3 / 2;

    if holeR_m > 0.00125
        fprintf(' WARNING: d=%.1f mm may overlap (spacing=2.5 mm)\n', holeDiameters_mm(ss));
    end

    holeLArr = holeCentres_m - holeR_m;
    holeRArr = holeCentres_m + holeR_m;

    fprintf(' d=%.1f mm ... ', holeDiameters_mm(ss));
    [res, rz, K, ~, peakJ] = solveWithHoles(holeLArr, holeRArr, p, true);
    fprintf('K=%.2f, peakJ=%.4f\n', K, peakJ);

    evalZ = p.d + 0.0005;
    [dist, Jline] = extractProfile(rz, p, evalZ);

    results(ss) = struct('diam_mm',holeDiameters_mm(ss), 'K',K, 'peakJ',peakJ, ...
        'res',res, 'rz',rz, 'dist',dist, 'Jline',Jline);
end

%% 6. Baseline profile
[distSolid, JlineSolid] = extractProfile(rzSolid, p, p.d + 0.0005);

```

```

%% 7. Results table
fprintf('\n %-10s %-8s %-12s %-8s %-8s\n', 'd(mm)', 'K', 'peakJ', 'dK(%)', 'dJ(%)');
fprintf(' %-10s %-8.2f %-12.4f\n', 'Solid', Ks, pJs);
for ss = 1:nSweep
    dK = (results(ss).K - Ks)/Ks*100;
    dJ = (results(ss).peakJ - pJs)/pJs*100;
    fprintf(' %-10.1f %-8.2f %-12.4f %--8.1f %--8.1f\n', ...
        results(ss).diam_mm, results(ss).K, results(ss).peakJ, dK, dJ);
end
fprintf('\n');

%% 8. Figures
fntSz = 10; fntName = 'Helvetica';
setAx = @(ax) set(ax, 'FontSize',fntSz, 'FontName',fntName, ...
    'TickDir','out', 'LineWidth',0.5, 'Box','on');

sweepCols = [linspace(0,0.85,nSweep); linspace(0.2,0.1,nSweep); linspace(0.7,0.1,nSweep)];

% Fig 1: All line profiles
figure('Units','cm','Position',[2 4 17.4 10],'Color','w','PaperPositionMode','auto');
plot(distSolid*100, JlineSolid, 'k--', 'LineWidth',2.0); hold on;
leg = {'Solid'};
for ss = 1:nSweep
    lw = 1.0; if results(ss).diam_mm==0.4, lw=2.2; end
    plot(results(ss).dist*100, results(ss).Jline, '-', 'Color',sweepCols(ss,:), 'LineWidth',lw);
    leg{end+1} = sprintf('d=%.1fmm', results(ss).diam_mm); %#ok<AGROW>
end
xline(-p.R_pad*100,':','Color',[.4 .4 .4],'LineWidth',0.7);
xline( p.R_pad*100,':','Color',[.4 .4 .4],'LineWidth',0.7); hold off;
xlabel('Distance from pad centre (cm)'); ylabel('|J| (A/m^2)');
title('Hole Diameter Sweep: Line Profiles');
legend(leg,'Location','southoutside','NumColumns',3,'FontSize',7,'Box','off');
xlim([-6 6]); setAx(gca); grid on;

% Fig 2: K and peakJ vs diameter
diams = [results.diam_mm]; Kvec = [results.K]; pJvec = [results.peakJ];
bIdx = find(diams==0.4);

figure('Units','cm','Position',[2 3 17.4 8],'Color','w','PaperPositionMode','auto');
sgtitle('Effect of Hole Diameter','FontSize',fntSz+1,'FontName',fntName,'FontWeight','bold');

subplot(1,2,1);
plot(diams, Kvec, 'ko-', 'MarkerFaceColor',[.2 .4 .8], 'MarkerSize',6, 'LineWidth',1.2);
hold on; yline(Ks, 'r--', 'LineWidth',1);
if ~isempty(bIdx), plot(0.4, Kvec(bIdx), 'rs', 'MarkerSize',12, 'MarkerFaceColor',[1 .6 .6], 'LineWidth',1.5); end
hold off;
xlabel('Hole diameter (mm)'); ylabel('K');
title('(a) Non-uniformity'); legend('Ring','Solid','0.4mm','Location','best','FontSize',7,'Box','off');
setAx(gca); grid on;

subplot(1,2,2);
plot(diams, pJvec, 'ko-', 'MarkerFaceColor',[.8 .3 .2], 'MarkerSize',6, 'LineWidth',1.2);
hold on; yline(pJs, 'r--', 'LineWidth',1);
if ~isempty(bIdx), plot(0.4, pJvec(bIdx), 'rs', 'MarkerSize',12, 'MarkerFaceColor',[1 .6 .6], 'LineWidth',1.5); end
hold off;
xlabel('Hole diameter (mm)'); ylabel('Peak |J| (A/m^2)');
title('(b) Peak tissue |J|'); legend('Ring','Solid','0.4mm','Location','best','FontSize',7,'Box','off');
setAx(gca); grid on;

```

% Fig 3: Full-domain |J| maps (solid vs 0.4mm)

```
if ~isempty(bIdx)
    th = linspace(0,2*pi,150);
    hx = p.heartX*100 + p.heartRadius*100*cos(th);
    hz = p.heartZ*100 + p.heartRadius*100*sin(th);
    Jall = [resSolid.J(:); results(bIdx).res.J(:)];
    clim = [0, prctile(Jall(~isnan(Jall)),99)];

    figure('Units','cm','Position',[2 2 17.4 8],'Color','w','PaperPositionMode','auto');
    sgtitle('|J| Maps: Solid vs Ring (0.4mm)','FontSize',fntSz+1,'FontName',fntName,'FontWeight','bold');

    for kk = 1:2
        ax = subplot(1,2,kk);
        if kk==1, dat=resSolid; lbl=sprintf('(a) Solid, K=%.2f',Ks);
        else, dat=results(bIdx).res; lbl=sprintf('(b) Ring 0.4mm, K=%.2f',results(bIdx).K); end
        imagesc([0 50],[0 25],dat.J); colormap(ax,jet(256)); caxis(clim);
        hold on; plot(hx,hz,'w-','LineWidth',1);
        plot(p.probeX*100,p.probeZ*100,'w+','MarkerSize',6,'LineWidth',1.2); hold off;
        xlabel('x (cm)'); ylabel('z (cm)'); title(lbl);
        axis equal tight; xlim([0 50]); ylim([0 25]);
        set(ax,'YDir','normal'); setAx(ax);
    end
    cb=colorbar('Position',[0.925 0.17 0.018 0.70]); ylabel(cb,'|J| (A/m^2)');
end
```

% Fig 4: Zoomed maps

```
if ~isempty(bIdx)
    zAll = [rzSolid.J(:); results(bIdx).rz.J(:)];
    zClim = [0, prctile(zAll(~isnan(zAll)),98)];

    figure('Units','cm','Position',[2 1 17.4 8],'Color','w','PaperPositionMode','auto');
    sgtitle('Zoomed |J| Under Electrode','FontSize',fntSz+1,'FontName',fntName,'FontWeight','bold');

    for kk = 1:2
        ax = subplot(1,2,kk);
        if kk==1, rz=rzSolid; lbl='(a) Solid';
        else, rz=results(bIdx).rz; lbl='(b) Ring 0.4mm'; end
        imagesc(rz.X(1,:)*100, rz.Z(:,1)*100, rz.J);
        colormap(ax,jet(256)); caxis(zClim);
        hold on; plot([p.pad1L p.pad1R]*100,[0 0],'w-','LineWidth',2); hold off;
        xlabel('x (cm)'); ylabel('z (cm)'); title(lbl);
        xlim([p.pad1L-0.005,p.pad1R+0.005]*100); ylim([0 2.5]);
        set(ax,'YDir','normal'); setAx(ax);
    end
    cb=colorbar('Position',[0.925 0.17 0.018 0.70]); ylabel(cb,'|J| (A/m^2)');
end
```

% Fig 5: Solid vs 0.4mm overlay

```
figure('Units','cm','Position',[2 0 12 8],'Color','w','PaperPositionMode','auto');
plot(distSolid*100, JlineSolid, 'k-', 'LineWidth',1.8); hold on;
if ~isempty(bIdx)
    plot(results(bIdx).dist*100, results(bIdx).Jline, '-','Color',[.8 .15 .15],'LineWidth',1.8);
end
for rr = 1:nRings
    xline(-ringRadii_m(rr)*100,':','Color',[.6 .6 .6],'LineWidth',0.5);
    xline( ringRadii_m(rr)*100,':','Color',[.6 .6 .6],'LineWidth',0.5);
end
xline(-p.R_pad*100,'-','Color',[.3 .3 .3],'LineWidth',0.8);
xline( p.R_pad*100,'-','Color',[.3 .3 .3],'LineWidth',0.8); hold off;
xlabel('Distance from pad centre (cm)'); ylabel('J_S (A/m^2)');
title('Solid vs Ring (d=0.4mm)');
```



```

legend('Solid','Ring (0.4mm)','Location','south','FontSize',8,'Box','off');
xlim([-6 6]); setAx(gca); grid on;

% Fig 6: % change
figure('Units','cm','Position',[2 0 12 8],'Color','w','PaperPositionMode','auto');
dK_pct = ([results.K]-Ks)/Ks*100;
dJ_pct = ([results.peakJ]-pJs)/pJs*100;
plot(diams,dK_pct,'o-','Color',[.2 .4 .8],'MarkerFaceColor',[.2 .4 .8],'MarkerSize',5,'LineWidth',1.2)
; hold on;
plot(diams,dJ_pct,'s-','Color',[.8 .3 .2],'MarkerFaceColor',[.8 .3 .2],'MarkerSize',5,'LineWidth',1.2)
;
yline(0,'k:','LineWidth',0.7);
if ~isempty(bIdx), xline(0.4,'--','Color',[.4 .4 .4],'LineWidth',0.8); end
hold off;
xlabel('Hole diameter (mm)'); ylabel('Change from solid (%)');
title('Performance vs Hole Diameter');
legend('\Delta K','\Deltapeak J','Location','best','FontSize',8,'Box','off');
setAx(gca); grid on;

%% 9. Save
save('ring_electrode_results.mat', ...
    'results','resSolid','rzSolid','Ks','pJs', ...
    'ringRadii_m','nHolesPerRing','ringDiameters_cm', ...
    'holeDiameters_mm','distSolid','JlineSolid','p');
fprintf('Results saved. Figures generated.\n');
end

%% Forward solver: discrete insulating holes
function [res, resZoom, K, JAtProbe, peakJtissue] = solveWithHoles(holeLArr, holeRArr, p, highRes)
    nHoles = length(holeLArr);
    useHoles = nHoles > 0;

    function tf = inHole(xi)
        if ~useHoles, tf = false;
        else, tf = any((xi >= holeLArr) & (xi <= holeRArr)); end
    end

    mdl = createpde();
    geometryFromEdges(mdl, p.dl);
    if highRes
        generateMesh(mdl,'Hmax',0.0012,'Hmin',0.0003,'GeometricOrder','linear');
    else
        generateMesh(mdl,'Hmax',0.002,'Hmin',0.0005,'GeometricOrder','linear');
    end

    function sig = condCoeff(region, ~)
        x = region.x; z = region.y;
        sig = p.sigma_tissue * ones(size(x));
        for ii = 1:numel(x)
            xi = x(ii); zi = z(ii);
            if (xi-p.heartX)^2+(zi-p.heartZ)^2 <= p.heartRadius^2
                sig(ii) = p.sigma_heart;
            elseif zi<=p.d && xi>=p.pad1L && xi<=p.pad1R
                if inHole(xi), sig(ii)=p.sigma_hole; else, sig(ii)=p.sigma_metal; end
            elseif zi>=p.domainHeight-p.d && xi>=p.pad2L && xi<=p.pad2R
                if inHole(xi), sig(ii)=p.sigma_hole; else, sig(ii)=p.sigma_metal; end
            end
        end
    end

    specifyCoefficients(mdl,'m',0,'d',0,'c',@condCoeff,'a',0,'f',0);
    applyBoundaryCondition(mdl,'dirichlet','Edge',p.elec1Edge,'u',0);

```

```

applyBoundaryCondition mdl, 'dirichlet', 'Edge', p.elec2Edge, 'u', p.Vapp);
for eID = setdiff(p.externalEdges, [p.elec1Edge, p.elec2Edge])
    applyBoundaryCondition mdl, 'neumann', 'Edge', eID, 'g', 0, 'q', 0);
end

result = solvepde(mdl);
nodeX = mdl.Mesh.Nodes(1,:); nodeZ = mdl.Mesh.Nodes(2,:);
[gradX, gradZ] = evaluateGradient(result, nodeX, nodeZ);

sigN = p.sigma_tissue*ones(size(nodeX));
for ii = 1:numel(nodeX)
    xi = nodeX(ii); zi = nodeZ(ii);
    if (xi-p.heartX)^2+(zi-p.heartZ)^2 <= p.heartRadius^2
        sigN(ii) = p.sigma_heart;
    elseif zi<=p.d && xi>=p.pad1L && xi<=p.pad1R
        if inHole(xi), sigN(ii)=p.sigma_hole; else, sigN(ii)=p.sigma_metal; end
    elseif zi>=p.domainHeight-p.d && xi>=p.pad2L && xi<=p.pad2R
        if inHole(xi), sigN(ii)=p.sigma_hole; else, sigN(ii)=p.sigma_metal; end
    end
end
Jmag = sqrt((-sigN.*gradX).^2 + (-sigN.*gradZ).^2);

% Interpolate onto grids
[gX,gZ] = meshgrid(linspace(0,p.domainWidth,250), linspace(0,p.domainHeight,125));
JG = griddata(nodeX,nodeZ,Jmag,gX,gZ);
VG = griddata(nodeX,nodeZ,result.NodalSolution,gX,gZ);
res = struct('J',JG, 'V',VG, 'X',gX, 'Z',gZ);

[zX,zZ] = meshgrid(linspace(p.pad1L-0.005,p.pad1R+0.005,500), linspace(0,0.025,200));
resZoom = struct('J',griddata(nodeX,nodeZ,Jmag,zX,zZ), ...
    'V',griddata(nodeX,nodeZ,result.NodalSolution,zX,zZ), 'X',zX, 'Z',zZ);

% K = Jmax/Jmin at 0.5mm below interface
evalZ = p.d + 0.0005;
[~,zi] = min(abs(gZ(:,1)-evalZ));
xm = (gX(1,:)>=p.pad1L) & (gX(1,:)<=p.pad1R);
Js = JG(zi,xm); Js = Js(~isnan(Js) & Js>0);
if ~isempty(Js), K = max(Js)/min(Js); else, K = NaN; end

JAtProbe = interp2(gX,gZ,JG,p.probeX,p.probeZ,'linear');
if isnan(JAtProbe)
    dd = (gX-p.probeX).^2+(gZ-p.probeZ).^2; dd(isnan(JG))=inf;
    [~,idx] = min(dd(:)); JAtProbe = JG(idx);
end

heartM = ((gX-p.heartX).^2+(gZ-p.heartZ).^2) <= p.heartRadius^2;
intfM = (gZ<=p.d) | (gZ>=p.domainHeight-p.d);
Jt = JG(~heartM & ~intfM & ~isnan(JG));
if ~isempty(Jt), peakJtissue = max(Jt); else, peakJtissue = NaN; end
end

%% Extract 1D profile
function [dist, Jline] = extractProfile(resZoom, p, evalZ)
    [~,zIdx] = min(abs(resZoom.Z(:,1)-evalZ));
    distFull = resZoom.X(1,:) - p.pad1X;
    Jrow = resZoom.J(zIdx,:);
    mask = abs(distFull) <= p.R_pad+0.005;
    dist = distFull(mask); Jline = Jrow(mask);
end

%% Build geometry

```

```

function [dl, externalEdges, elec1Edge, elec2Edge] = buildRingGeometry(p)
    vertices = [0,0; p.pad1L,0; p.pad1R,0; p.domainWidth,0;
                p.domainWidth,p.domainHeight; p.pad2R,p.domainHeight;
                p.pad2L,p.domainHeight; 0,p.domainHeight];
    nv = size(vertices,1);
    pgon = [2;nv;vertices(:,1);vertices(:,2)];
    circ = [1;p.heartX;p.heartZ;p.heartRadius;zeros(6,1)];
    maxLen = max(length(pgon),length(circ));
    pgon(end+1:maxLen)=0; circ(end+1:maxLen)=0;

    [dl,~] = decsg([pgon,circ], 'P1+C1', char('P1','C1'));

    numEdges = size(dl,2); externalEdges = [];
    for eID = 1:numEdges
        if dl(6,eID)==0 || dl(7,eID)==0, externalEdges(end+1)=eID; end %#ok<AGROW>
    end

    elec1Edge=2; elec2Edge=6; tol=1e-6;
    for eID = externalEdges
        mx=(dl(2,eID)+dl(3,eID))/2; my=(dl(4,eID)+dl(5,eID))/2;
        if abs(my)<tol && mx>p.pad1L-tol && mx<p.pad1R+tol, elec1Edge=eID; end
        if abs(my-p.domainHeight)<tol && mx>p.pad2L-tol && mx<p.pad2R+tol, elec2Edge=eID; end
    end
    fprintf('Geometry: %d edges, elec1=%d, elec2=%d\n\n', numEdges, elec1Edge, elec2Edge);
end

```

6.1.3 Back Pad Length Sweep

```

function sweepBackPadLength_only()
% Sweeps back pad length (bottom electrode width) while holding front pad
% constant at 10 cm. Outputs: JedgeMetric, Jheart, Reff, K, Itotal.

% Fixed parameters
rhoCenter = 20;
d = 0.001;
Vapp = 5;
pad1X = 0.25;
pad2X = 0.25;
rhoRing = 200;
ringW = 0.004;
padWFront = 0.10;

padWBackVals = [0.06 0.08 0.10 0.12 0.14 0.16 0.20 0.24];

% Domain and tissue configuration
cfg = struct();
cfg.domainWidth = 0.50;
cfg.domainThickness = 0.25;
cfg.heartRadius = 0.09;
cfg.heartCenter = struct('x', 0.25, 'z', 0.05 + 0.09);
cfg.tissueConductivity = 0.3;
cfg.heartConductivity = 0.7;
cfg.meshMaxSize = 0.001;

gridParams = struct();
gridParams.xRange = [0 cfg.domainWidth];
gridParams.zRange = [0 cfg.domainThickness];
gridParams.resX = 220;
gridParams.resZ = 120;

% Metric parameters

```

```

delta = 0.0005;
edgePercentile = 95;
heartWallThickness = 0.005;
skinZBand = 0.002;
edgeStripW = 0.003;

% Storage
nP = numel(padWBackVals);
JedgeMetric = nan(nP,1);
Jheart = nan(nP,1);
Reff = nan(nP,1);
Kvals = nan(nP,1);
Itotal = nan(nP,1);

zCenter = rhoCenter * d;
zRing = rhoRing * d;

if ringW >= padWFront/2
    error('Front pad fixed at %.1f mm but ringW >= padWFront/2 (ringW=%.2f mm).', ...
        padWFront*1e3, ringW*1e3);
end

fprintf('\nSweeping back pad width. Front pad held at %.1f mm, rhoRing=%g Ohm*m, ringW=%.2f mm\n\n', ...
    padWFront*1e3, rhoRing, ringW*1e3);

for p = 1:nP
    padWBack = padWBackVals(p);

    if ringW >= padWBack/2
        warning('Skipping back padW=%.1f mm: ringW >= padWBack/2.', padWBack*1e3);
        continue;
    end

    rThresholdBack = padWBack/2 - ringW;
    rThresholdFront = padWFront/2 - ringW;

    impedanceBack = @(r) zCenter + (zRing - zCenter) .* 0.5 .* (1 + tanh((r - rThresholdBack) ./
        delta));
    impedanceFront = @(r) zCenter + (zRing - zCenter) .* 0.5 .* (1 + tanh((r - rThresholdFront) ./
        delta));

    % Bottom/back electrode (0 V, swept width)
    elec1 = struct();
    elec1.position = pad1X;
    elec1.width = padWBack;
    elec1.edge = 'bottom';
    elec1.voltage = 0;
    elec1.interfaceThickness = d;
    elec1.impedanceFunc = impedanceBack;

    % Top/front electrode (Vapp, fixed width)
    elec2 = struct();
    elec2.position = pad2X;
    elec2.width = padWFront;
    elec2.edge = 'top';
    elec2.voltage = Vapp;
    elec2.interfaceThickness = d;
    elec2.impedanceFunc = impedanceFront;

    cfg.electrode1 = elec1;
    cfg.electrode2 = elec2;

```

```

    [model, geom] = buildThoraxModelWithInterface(cfg);
    dirichletBoundaryConditions(model, geom);
    out = solveDirichletProblem(model, geom, gridParams);

    X = out.gridX;
    Z = out.gridZ;
    J = out.JGrid;

    JedgeMetric(p) = edgeHotspotFromGrid(X, Z, J, pad1X, pad2X, padWBack, padWFront, skinZBand,
        edgeStripW, edgePercentile);
    Jheart(p) = heartWallMeanFromGrid(X, Z, J, cfg.heartCenter.x, cfg.heartCenter.z, cfg.
        heartRadius, heartWallThickness);
    Itotal(p) = out.totalCurrent;
    Reff(p) = out.effectiveResistance;
    Kvals(p) = out.nonUniformityCoeff;

    fprintf('backPadW=%.1fmm: Jedge(p%d)=%.4g, Jheart=%.4g, Itotal=%.4g, Reff=%.4g, K=%.4g\n', ...
        padWBack*1e3, edgePercentile, JedgeMetric(p), Jheart(p), Itotal(p), Reff(p), Kvals(p));
end

% Plotting
set(groot, 'defaultAxesFontName', 'Arial');
set(groot, 'defaultAxesFontSize', 12);
set(groot, 'defaultLineLineWidth', 2);
set(groot, 'defaultFigureColor', 'w');

xmm = padWBackVals * 1e3;
ftitle = sprintf('(Front = %.0f mm)', padWFront*1e3);

plotSweep(xmm, JedgeMetric, 'Back Pad Length (mm)', sprintf('Edge Hotspot |J| (p%d)',
    edgePercentile), ...
    ['Edge Hotspot vs Back Pad Length ' ftitle], [0 0.45 0.74]);

plotSweep(xmm, Jheart, 'Back Pad Length (mm)', 'Mean |J| on Heart Wall', ...
    ['Heart Wall Mean Current Density ' ftitle], [0.85 0.33 0.1]);

plotSweep(xmm, Itotal, 'Back Pad Length (mm)', 'Total Current (A)', ...
    ['Total Current vs Back Pad Length ' ftitle], [0.47 0.67 0.19]);

plotSweep(xmm, Reff, 'Back Pad Length (mm)', 'Effective Resistance (Ohms)', ...
    ['Effective Resistance vs Back Pad Length ' ftitle], [0.49 0.18 0.56]);

plotSweep(xmm, Kvals, 'Back Pad Length (mm)', 'Non-Uniformity Coefficient', ...
    ['Non-Uniformity vs Back Pad Length ' ftitle], [0.3 0.3 0.3]);
end

function plotSweep(x, y, xlab, ylab, ttl, color)
    figure('Color', 'w');
    plot(x, y, 'o-', 'MarkerSize', 6, 'MarkerFaceColor', color);
    box on; grid off;
    xlabel(xlab, 'FontWeight', 'bold');
    ylabel(ylab, 'FontWeight', 'bold');
    title(ttl, 'FontWeight', 'bold');
    set(gca, 'LineWidth', 1.2);
end

function Jm = edgeHotspotFromGrid(X, Z, J, pad1X, pad2X, padW1, padW2, skinZBand, edgeStripW, pct1)
% Returns percentile edge hotspot |J| sampled near pad edges on skin boundaries.
    if isempty(X) || isempty(Z) || isempty(J)
        Jm = NaN;
        warning('Empty grid arrays for edge metric.');
```

```

Jmag = abs(J);
zMin = min(Z(:));
zMax = max(Z(:));

onBottom = abs(Z - zMin) <= skinZBand;
onTop = abs(Z - zMax) <= skinZBand;

x1L = pad1X - padW1/2; x1R = pad1X + padW1/2;
x2L = pad2X - padW2/2; x2R = pad2X + padW2/2;

inPad1 = (X >= x1L) & (X <= x1R);
inPad2 = (X >= x2L) & (X <= x2R);

edge1 = (abs(X - x1L) <= edgeStripW) | (abs(X - x1R) <= edgeStripW);
edge2 = (abs(X - x2L) <= edgeStripW) | (abs(X - x2R) <= edgeStripW);

samples = [ Jmag(onBottom & inPad1 & edge1); ...
            Jmag(onTop & inPad2 & edge2) ];
samples = samples(isfinite(samples));

if isempty(samples)
    Jm = NaN;
    warning('No edge samples found. Try increasing skinZBand or edgeStripW.');
```

```

    return;
end

Jm = prctile(samples, pctl);
end

function Jh = heartWallMeanFromGrid(X, Z, J, hx, hz, R, t)
% Returns mean |J| over the heart wall annulus: R-t <= r <= R.
    if isempty(X) || isempty(Z) || isempty(J)
        Jh = NaN;
        warning('Empty grid arrays for heart metric.');
```

```

        return;
    end

    r = sqrt((X - hx).^2 + (Z - hz).^2);
    vals = abs(J((r >= R - t) & (r <= R)));
    vals = vals(isfinite(vals));

    if isempty(vals)
        Jh = NaN;
        warning('No heart-wall samples found. Check heartCenter/heartRadius vs domain/grid.');
```

```

        return;
    end

    Jh = mean(vals);
end

```